



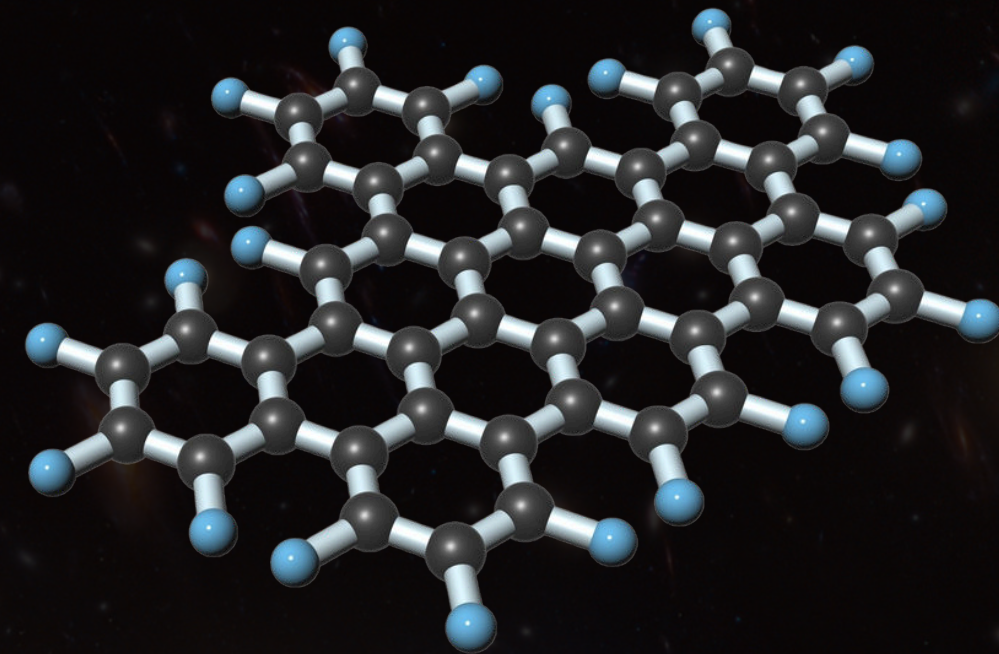
Galaxy clusters environmental effects on $\text{PAH}_{3.3}$ emission

Pietro Benotto



P
A
H

Polycyclic Aromatic Hydrocarbon

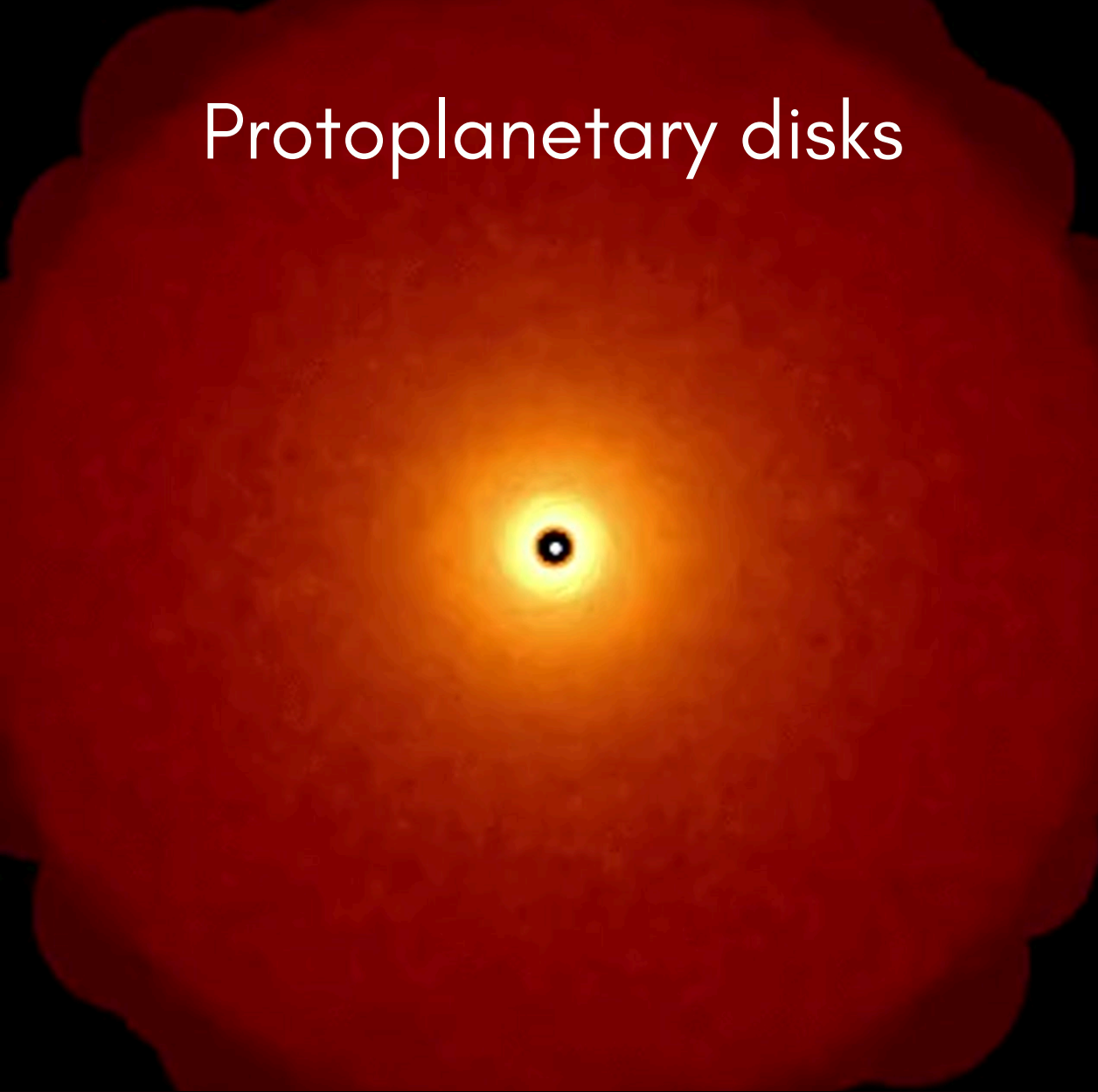


Polycyclic **A**romatic **H**ydrocarbon

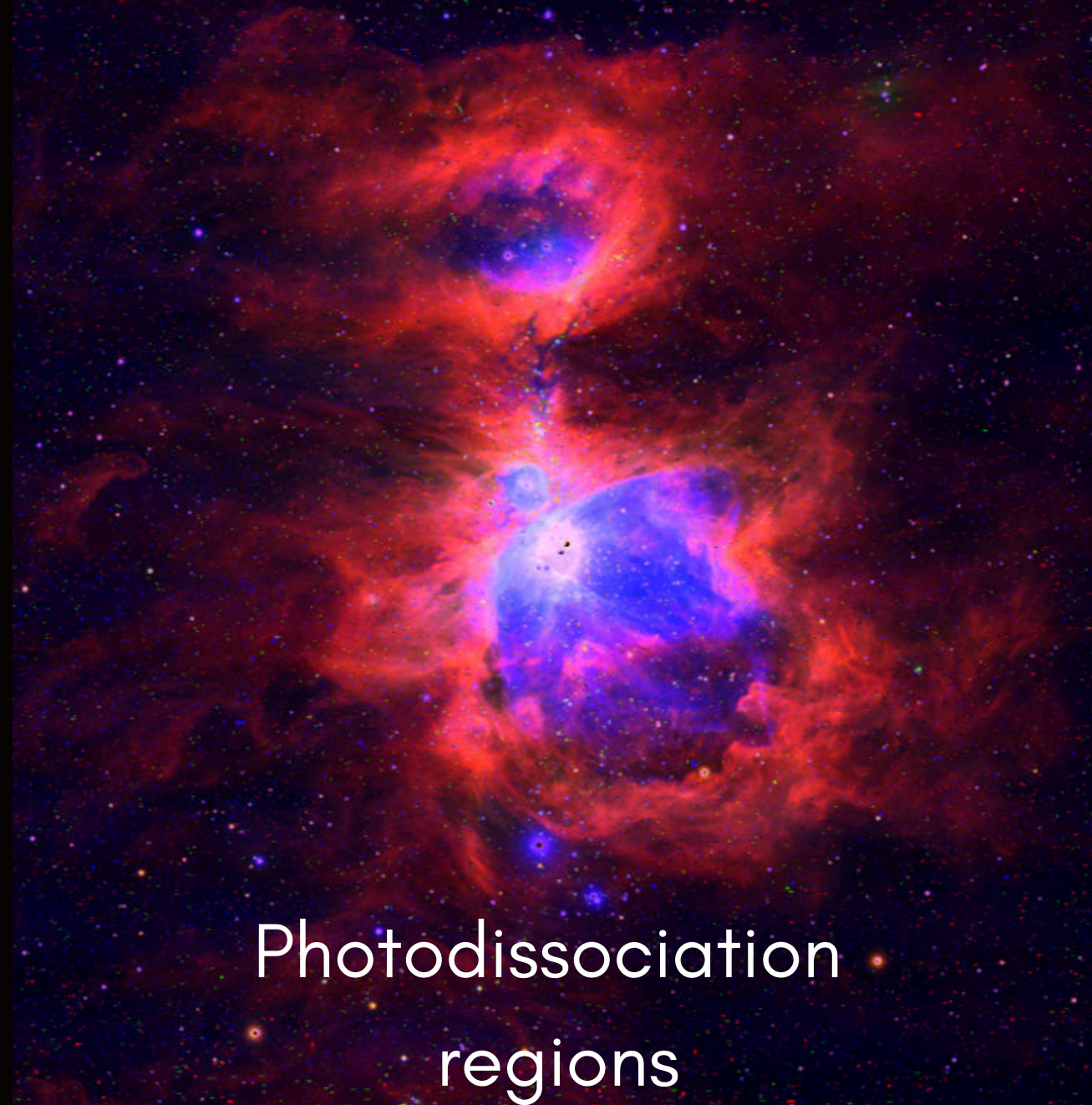
Building block:
Aromatic rings

Many
Carbon atoms

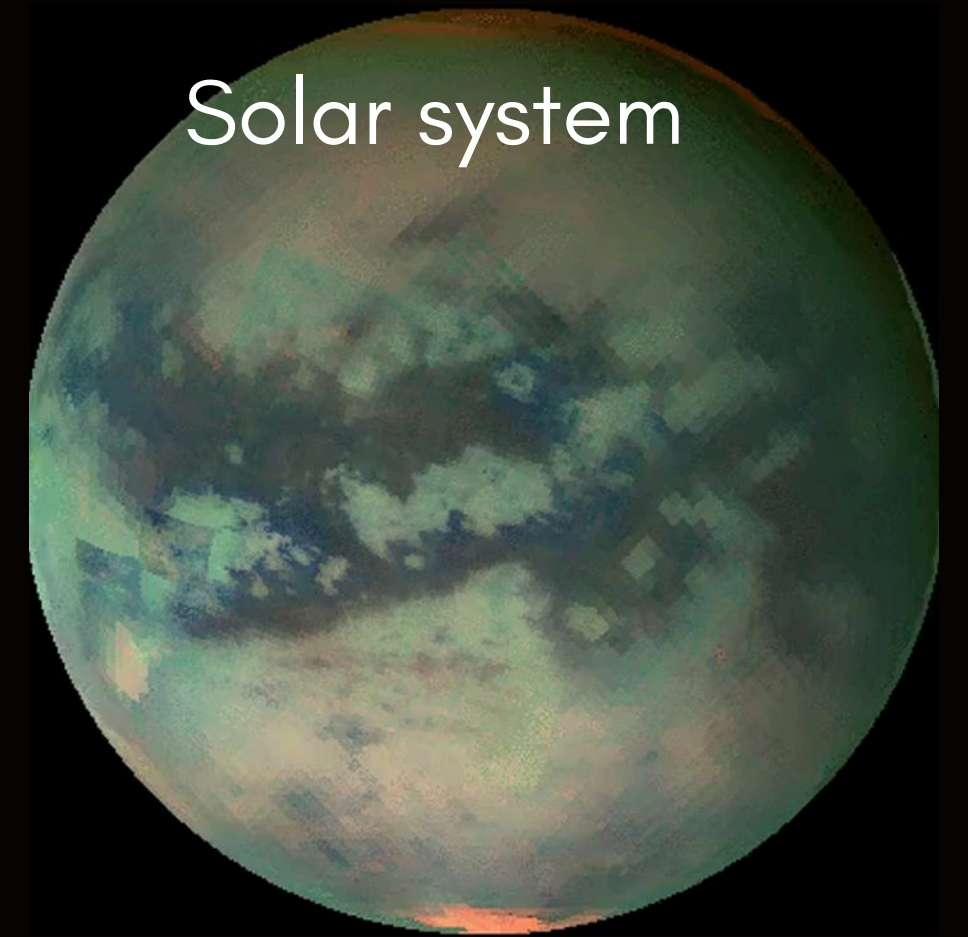
Protoplanetary disks



Photodissociation
regions



Solar system

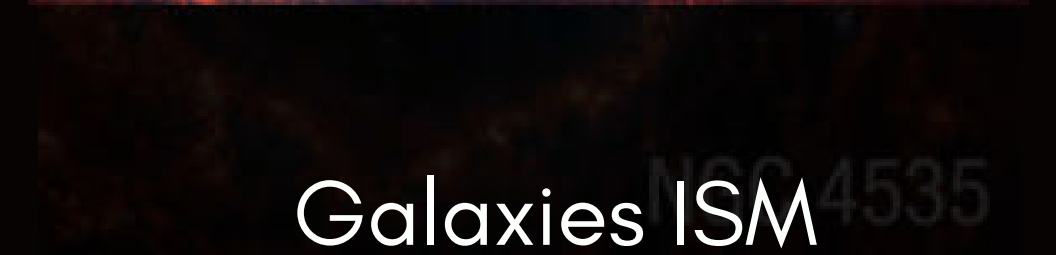


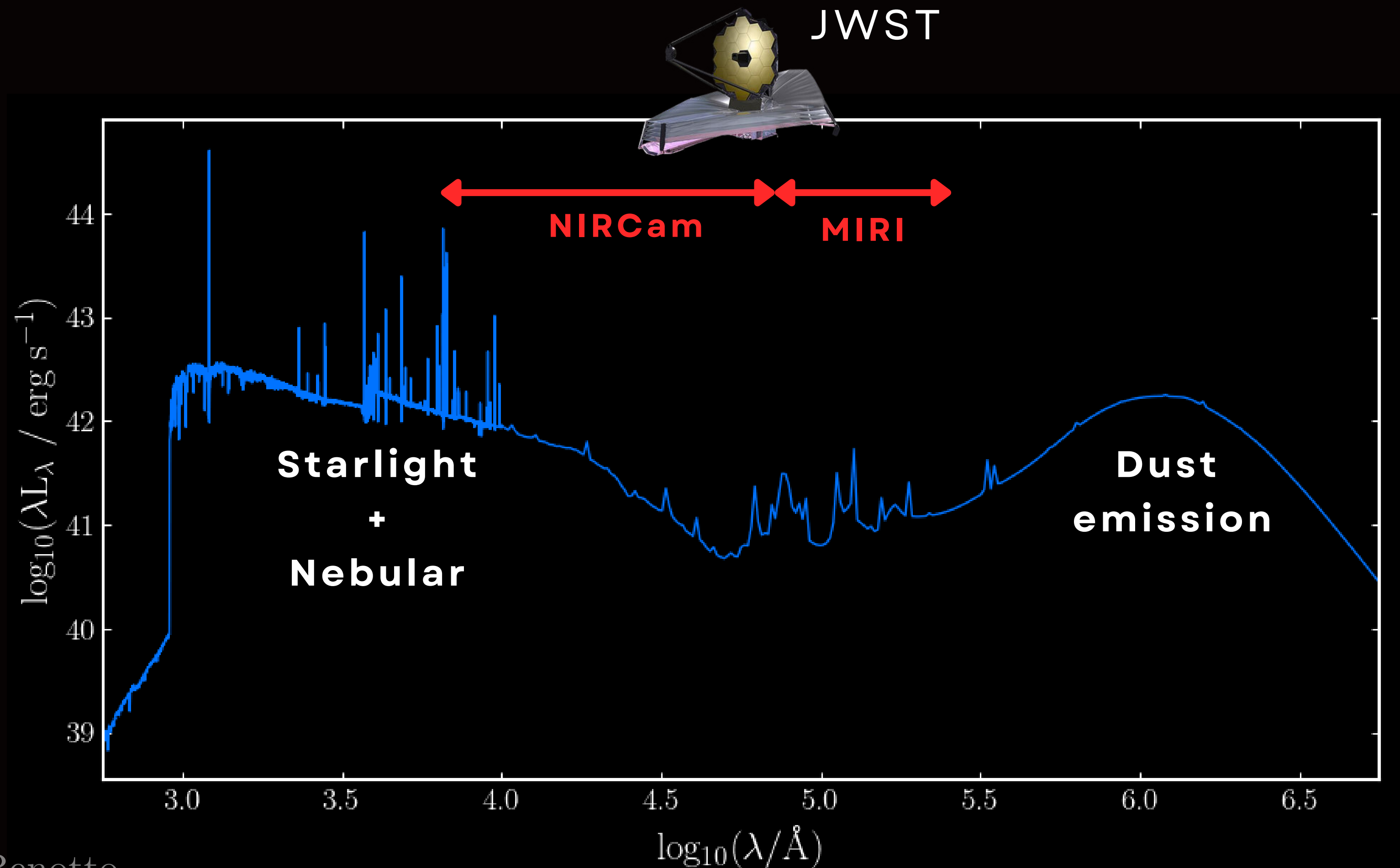
Milky way

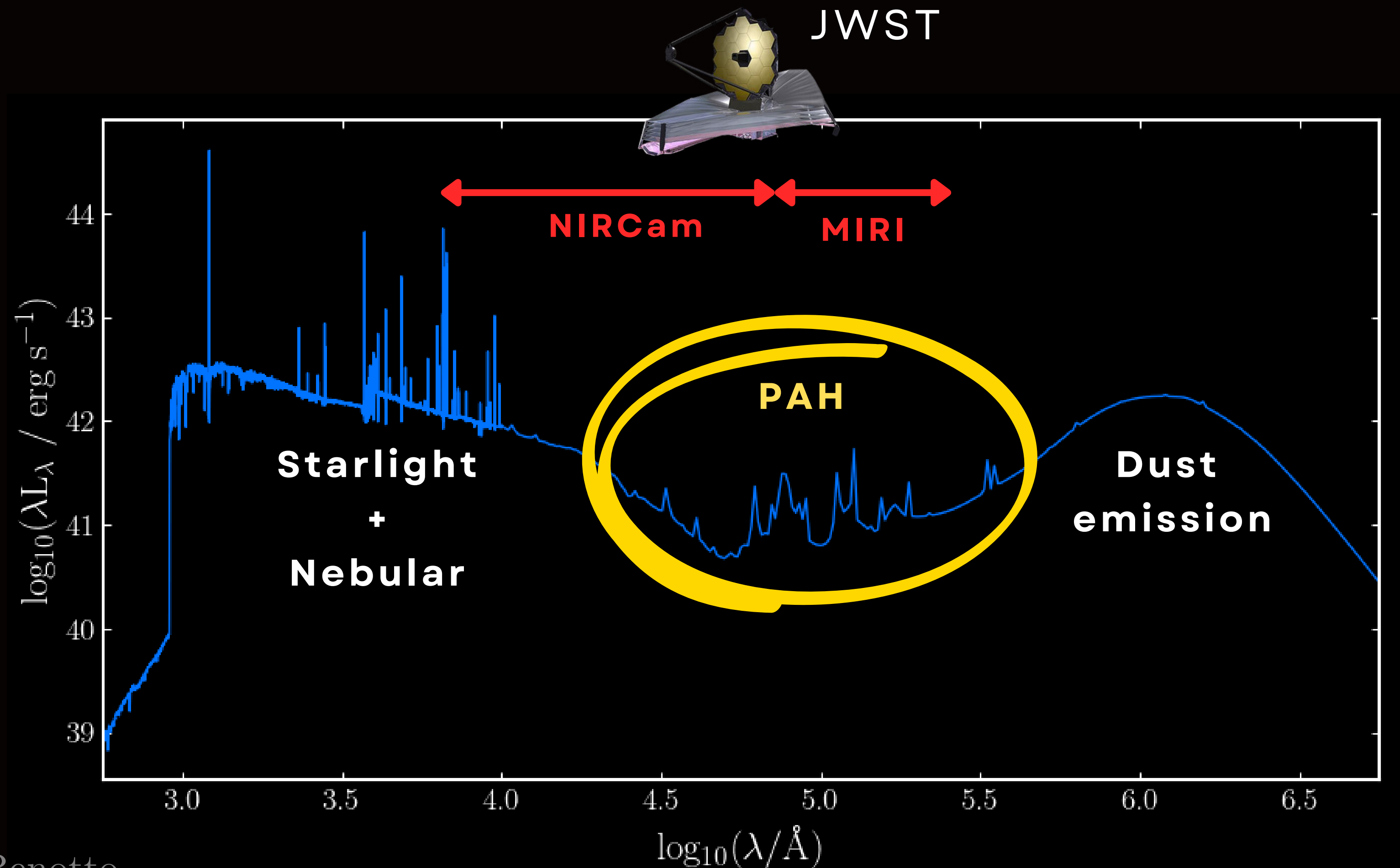
NGC 1433

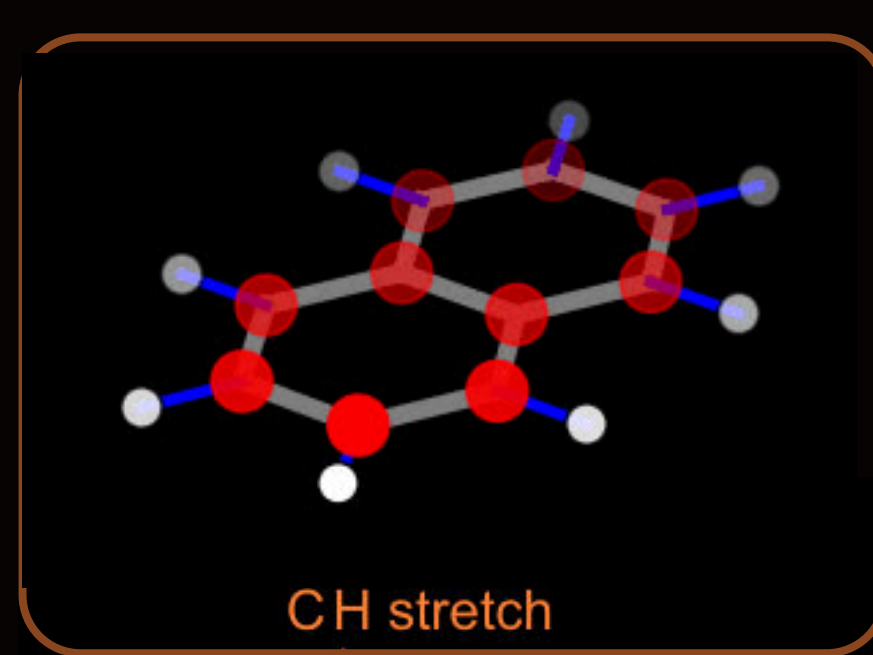
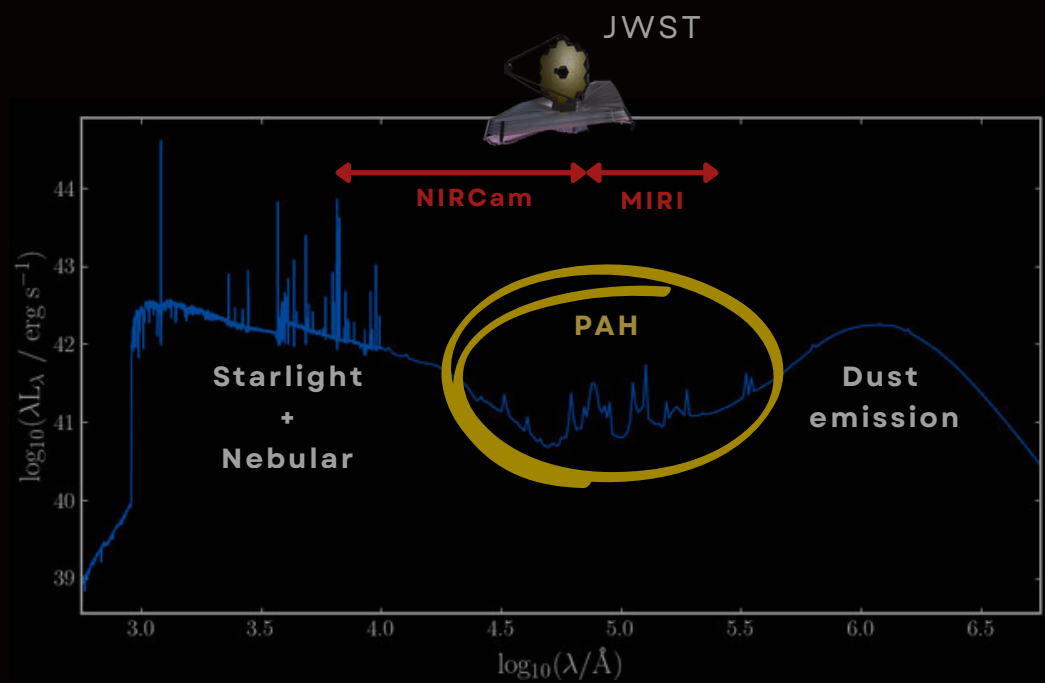


Galaxies ISM

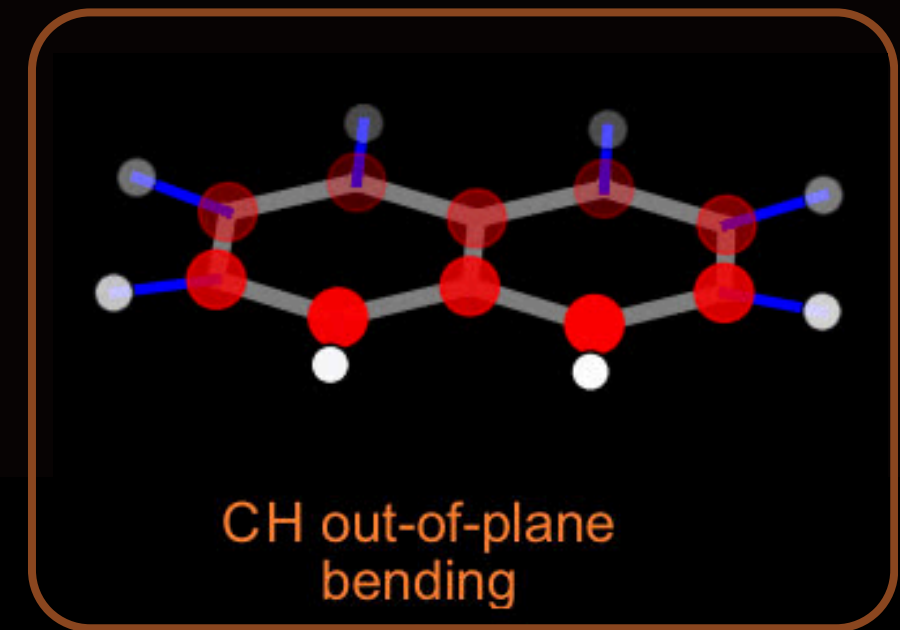




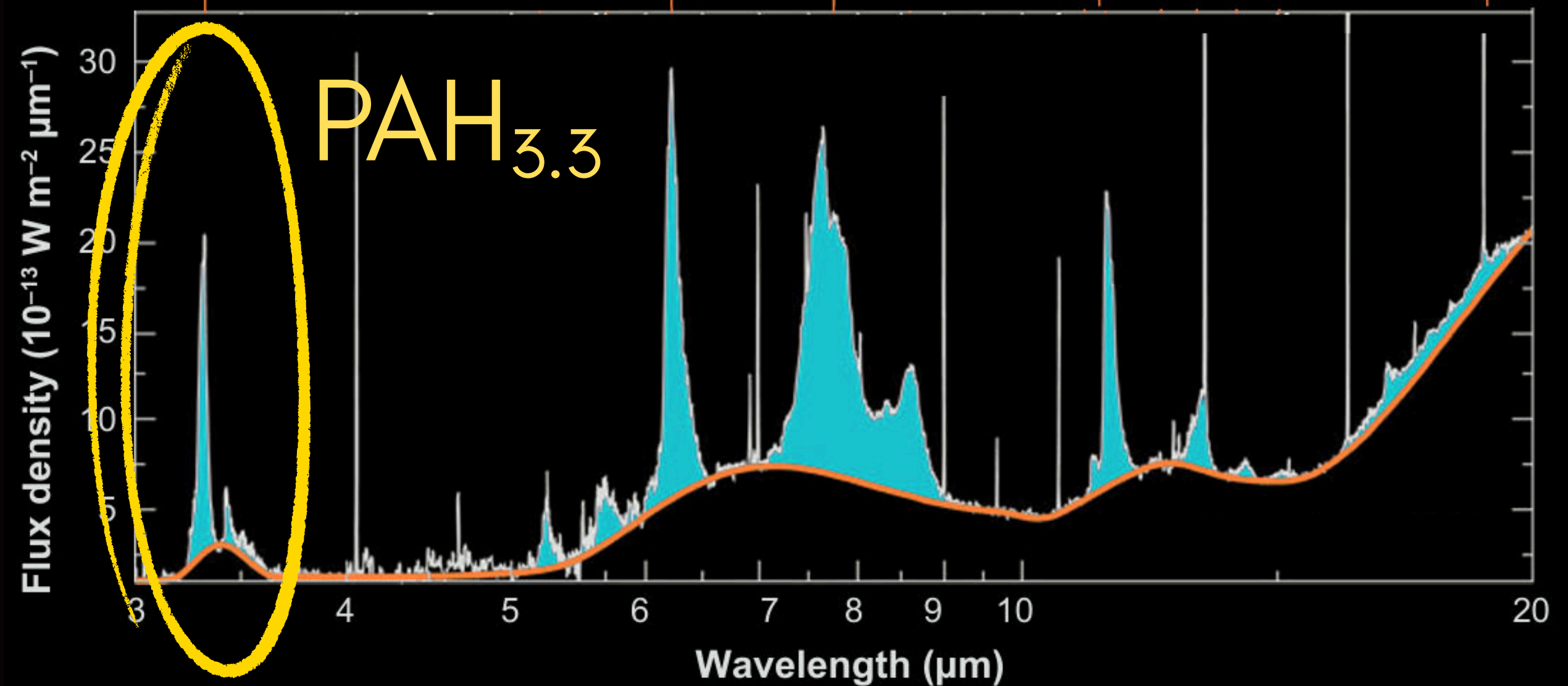


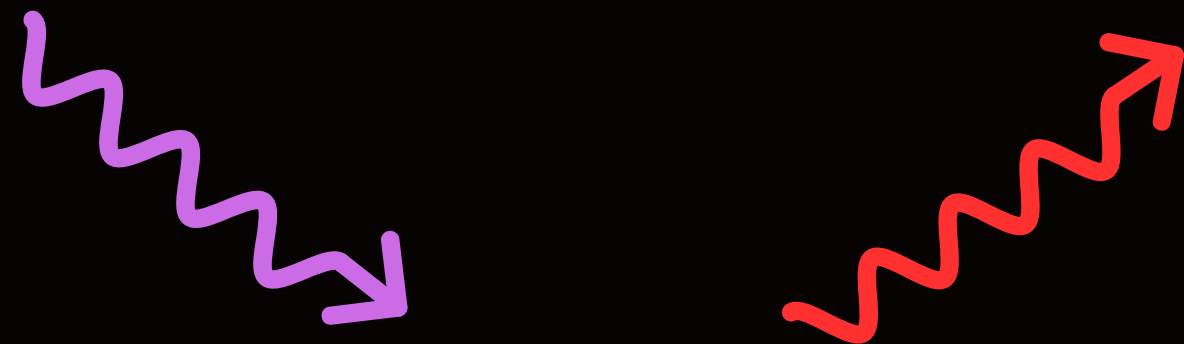


CC stretch

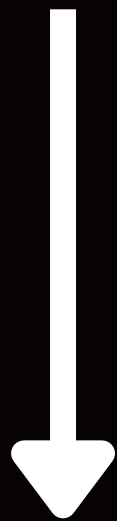


CC stretch

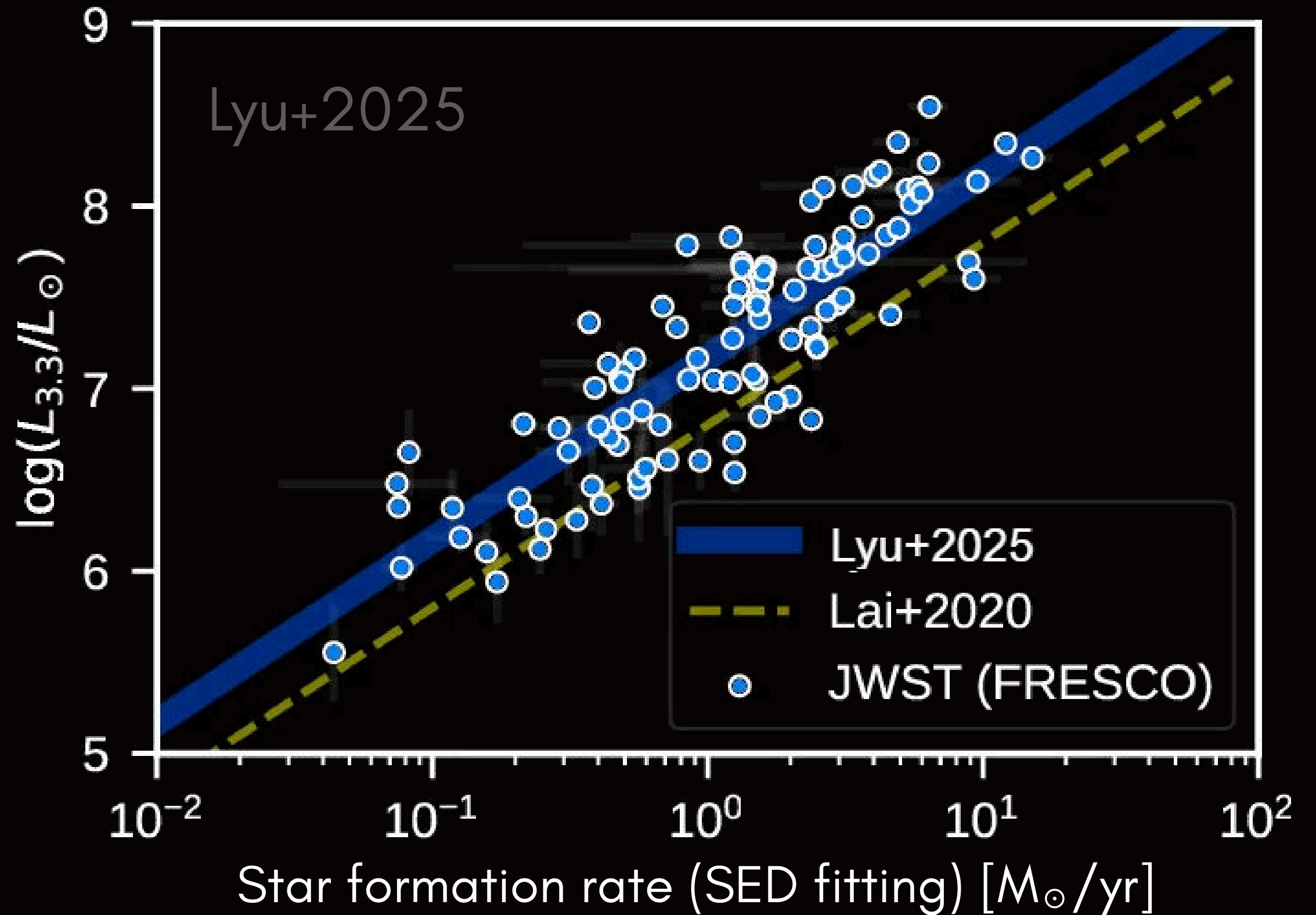
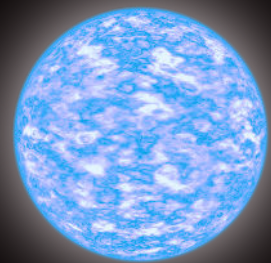




PAH molecules absorb UV photons and re-emit mid-infrared photons

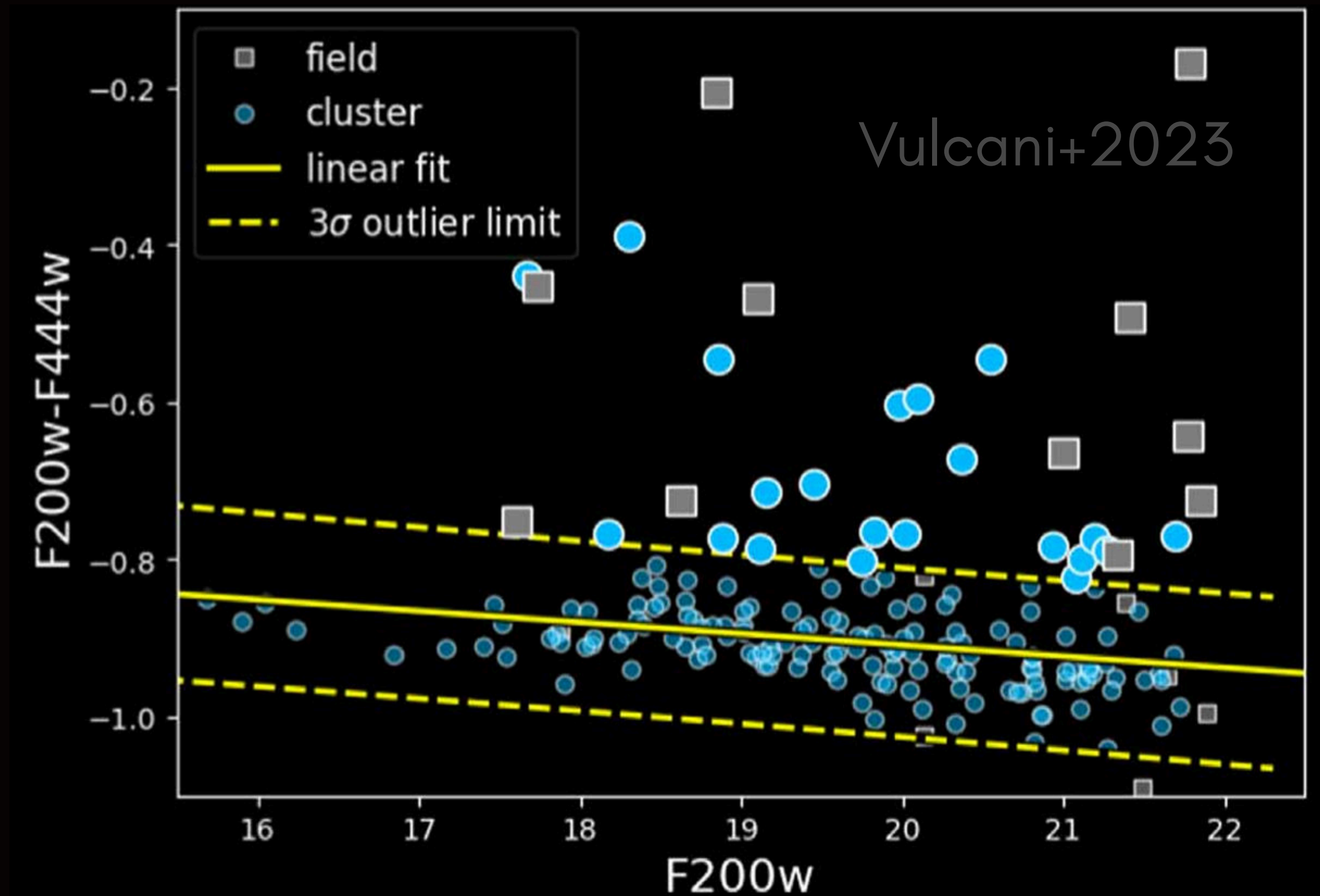


The PAH emission can be used as a **star formation tracer**



Red excess galaxies

(Vulcani+2023)



**Red excess
galaxies**

(Vulcani+2023)



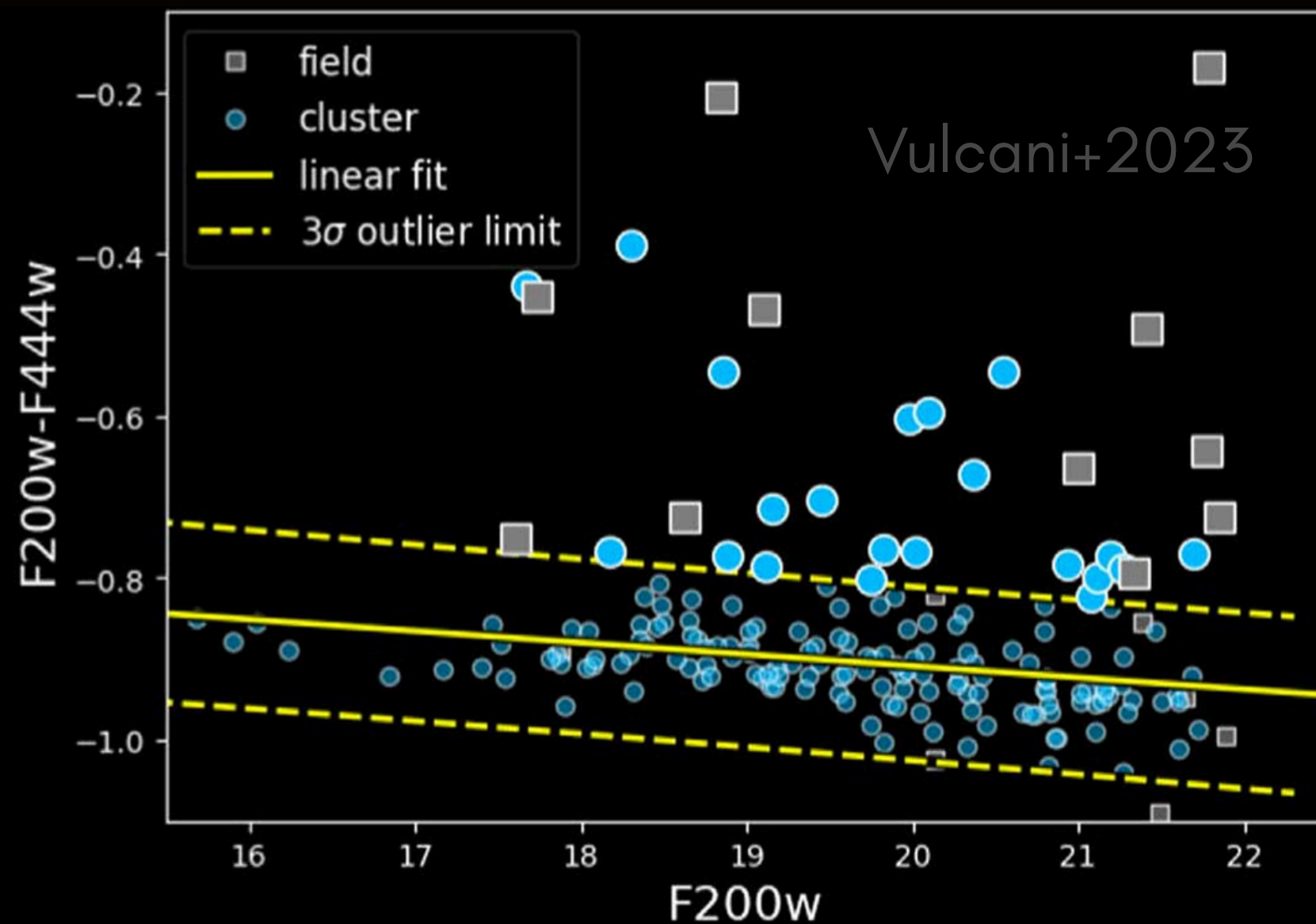
**PAH in galaxy
clusters**

(Vulcani+2025)



85% of red excess galaxies
are PAH emitters

Strong PAH emitters are
morphologically disturbed
galaxies



**Red excess
galaxies**

(Vulcani+2023)



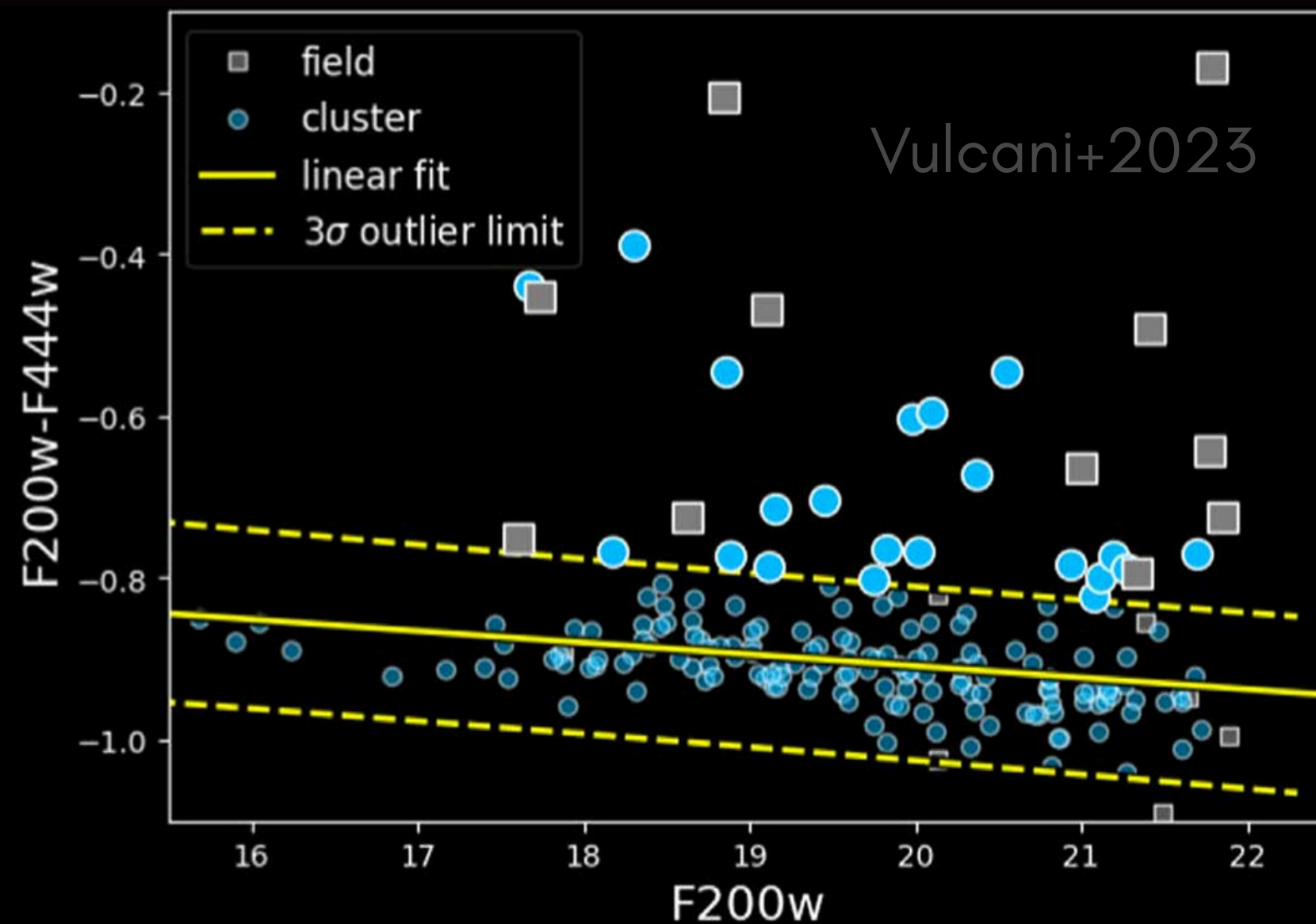
**PAH in galaxy
clusters**

(Vulcani+2025)



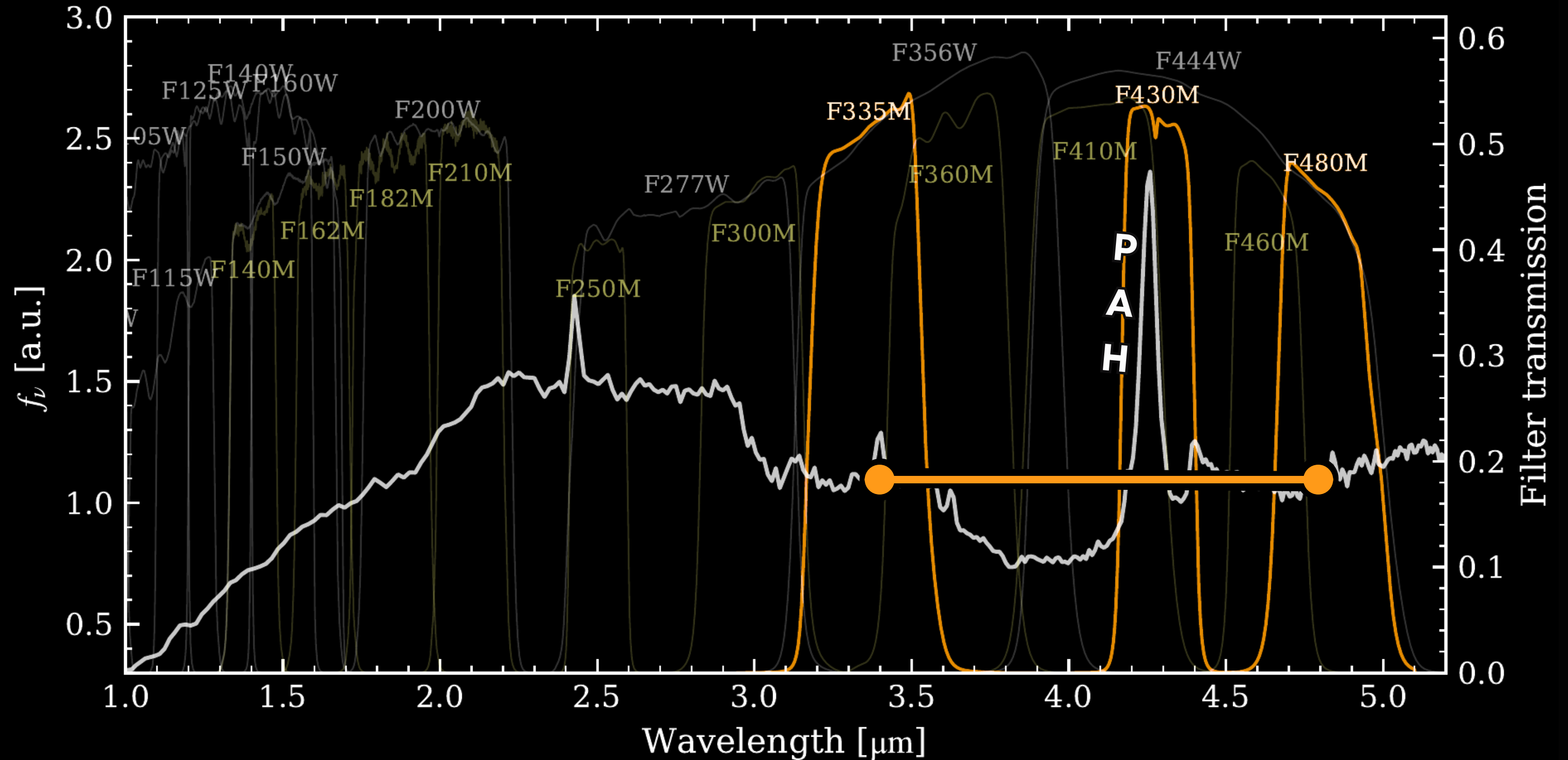
85% of red excess galaxies
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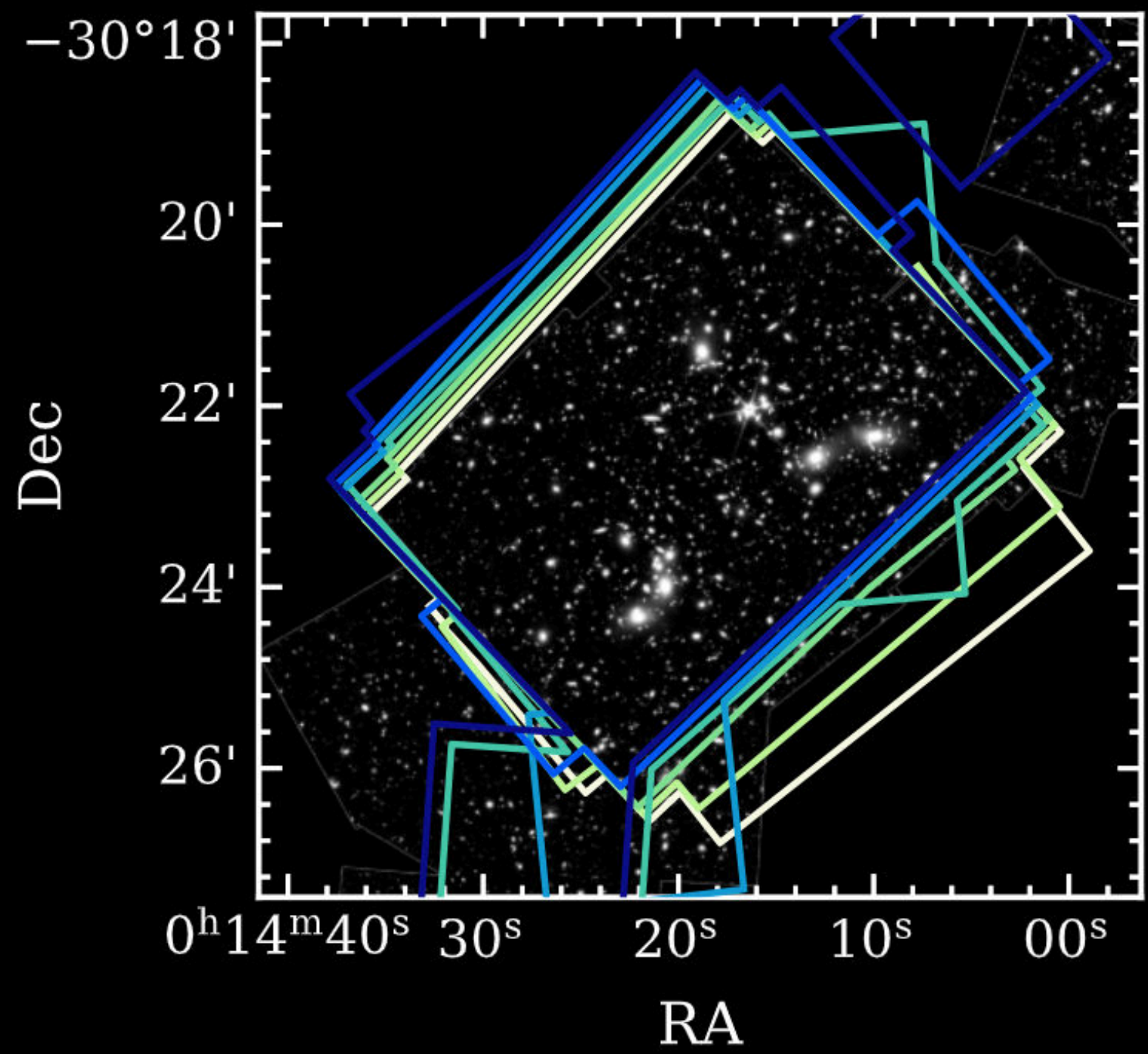
Strong PAH emitters are
morphologically disturbed
galaxies



**How does the cluster
environment affect
the PAH_{3.3} emission?**

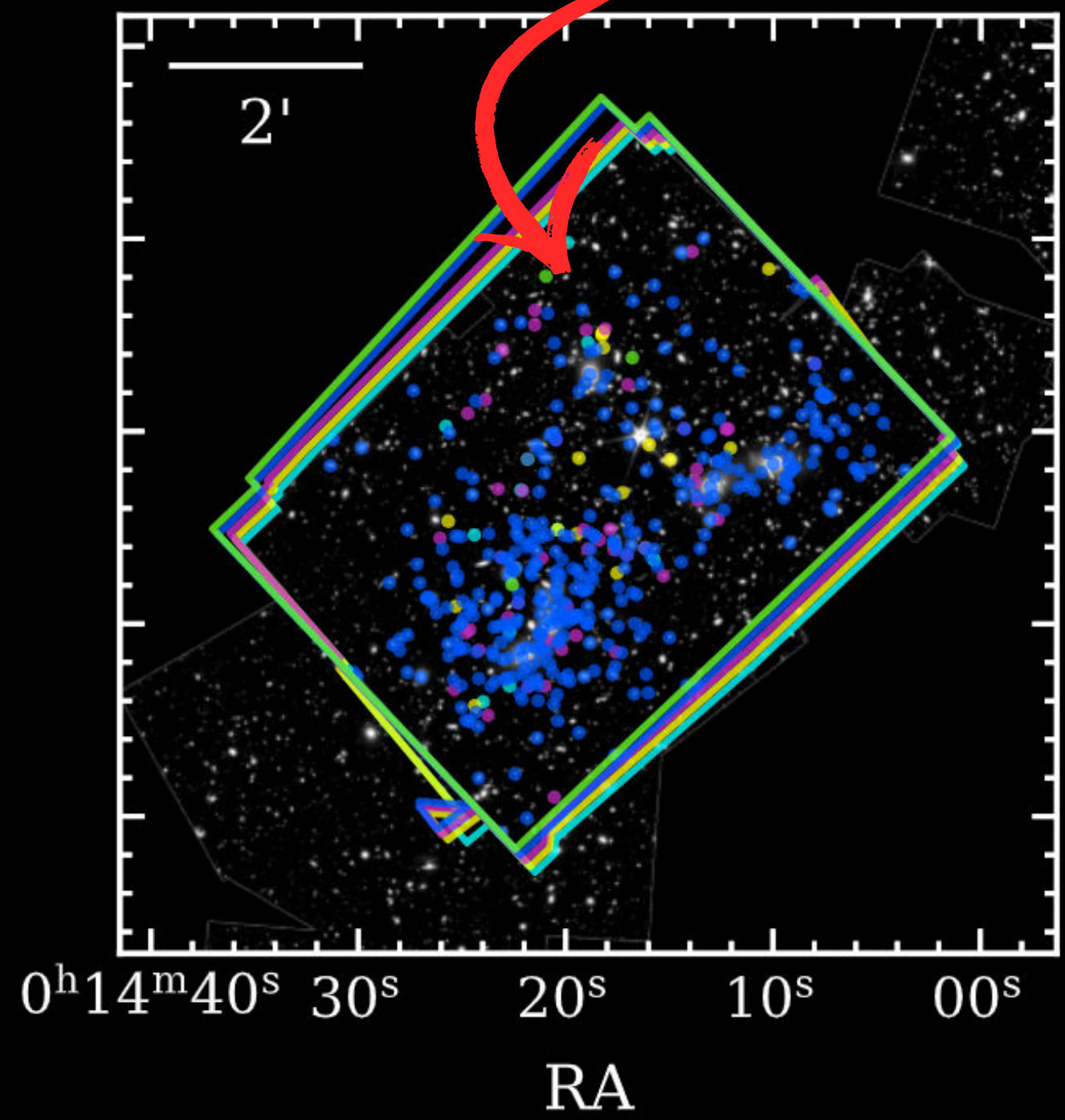
PAH flux extraction





Filter footprints

- | | |
|---|---|
| F300M | F410M |
| F335M | F430M |
| F360M | F460M |
| | F480M |

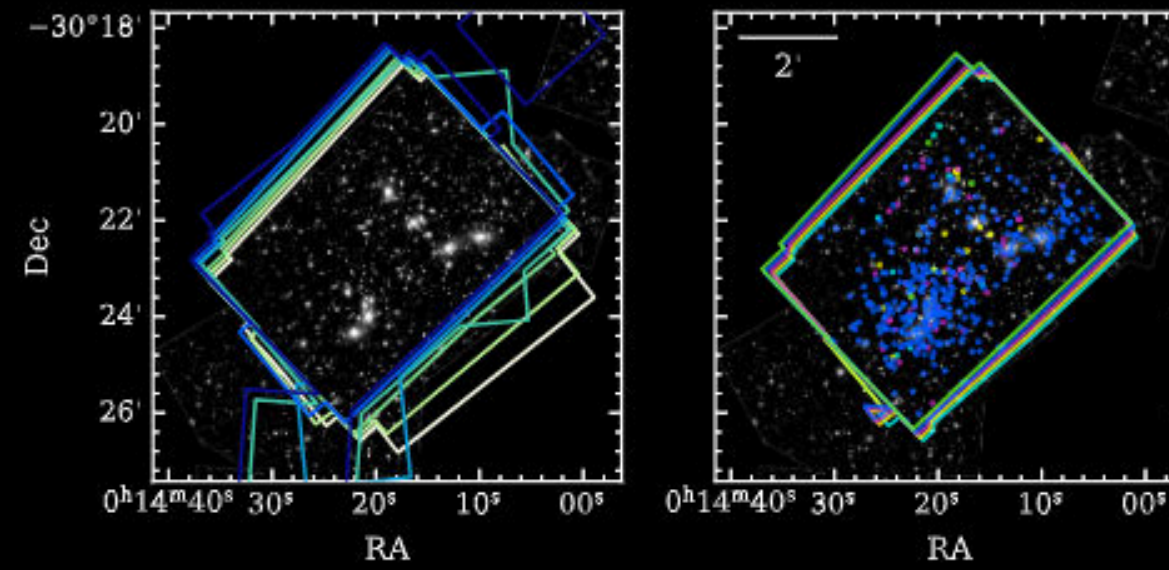
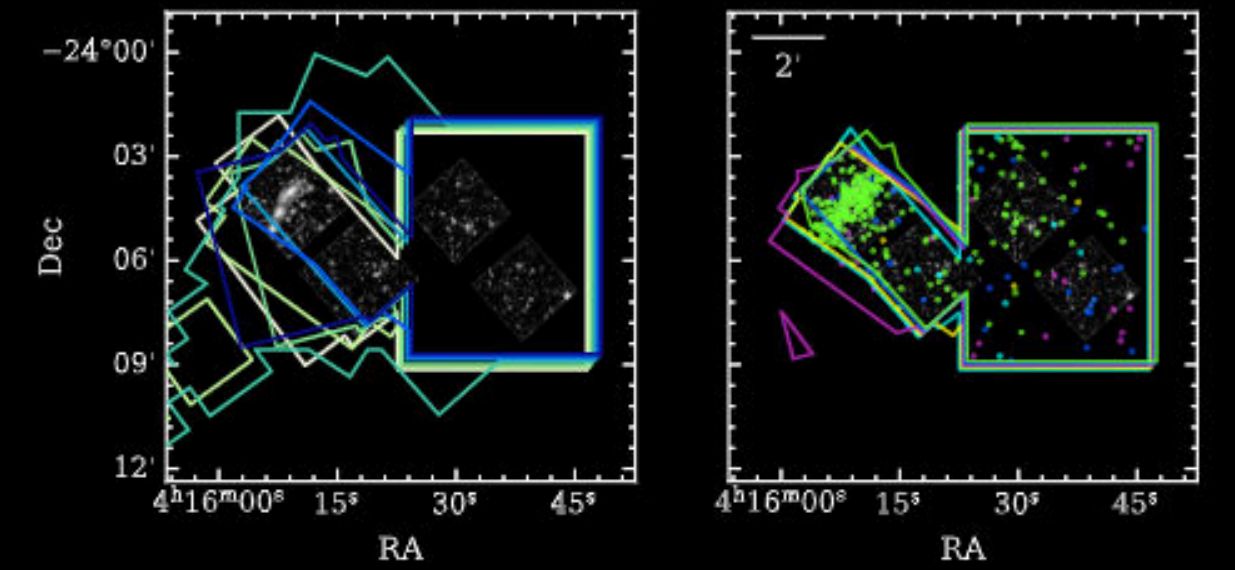
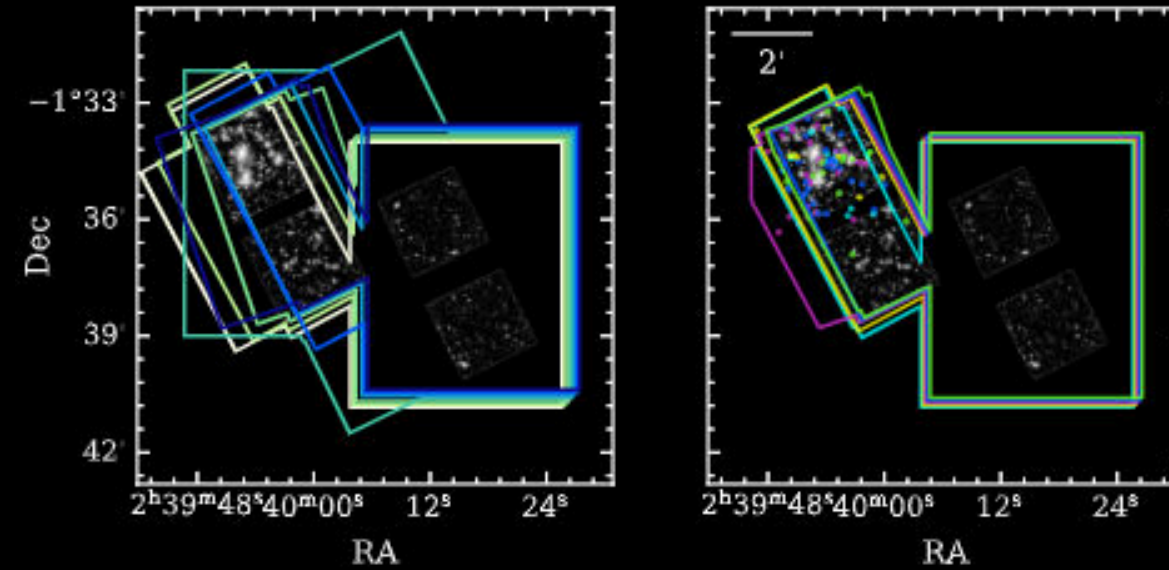
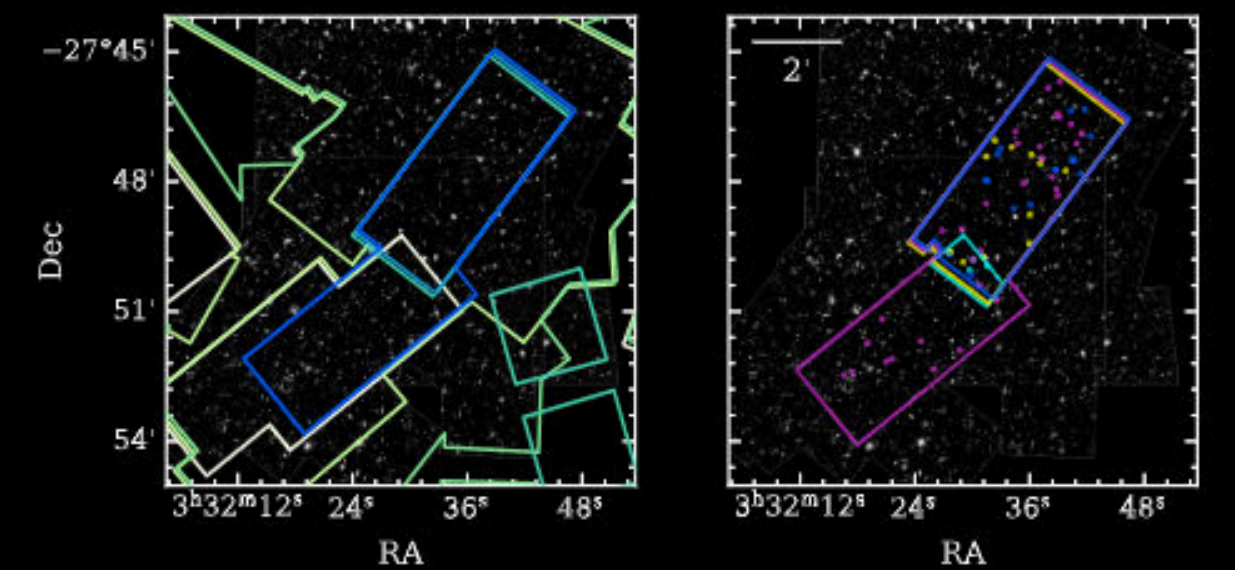
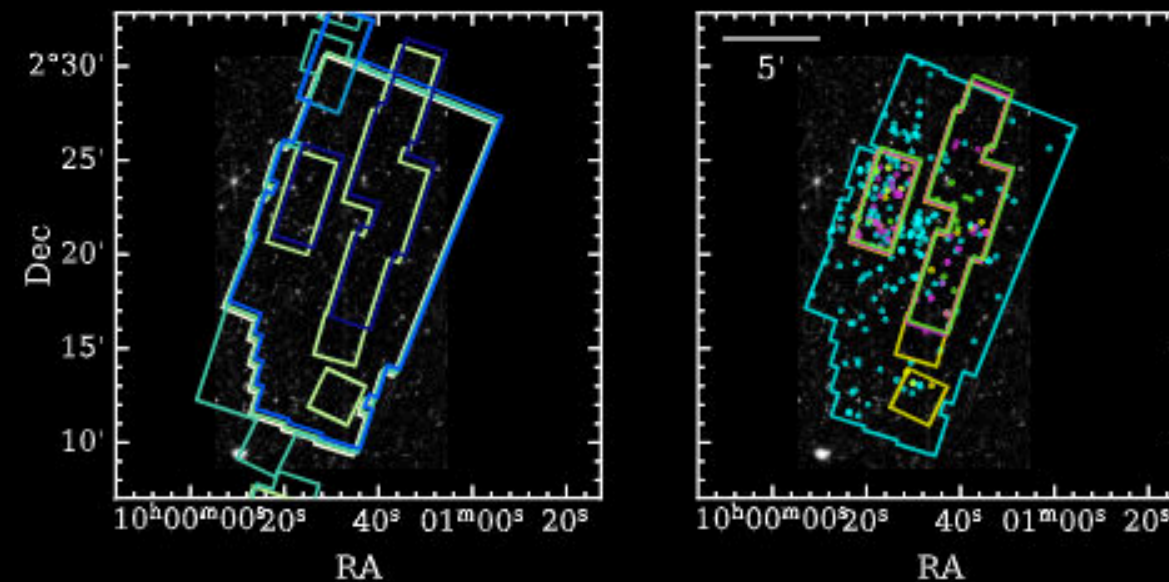
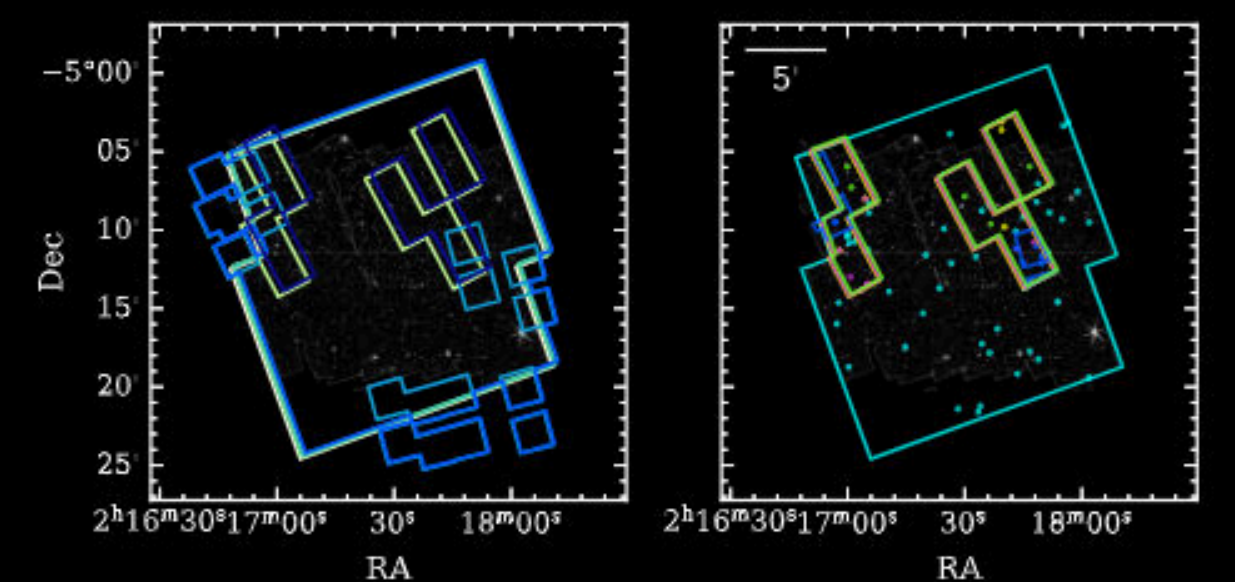
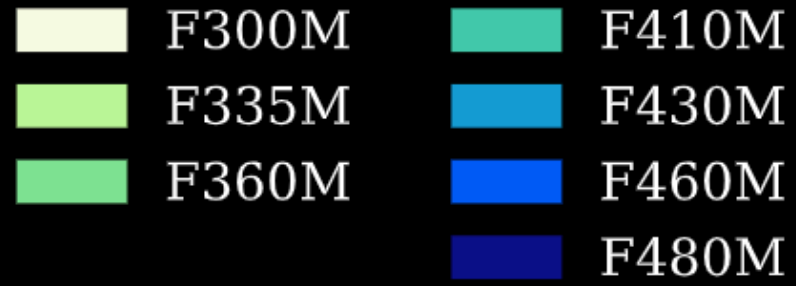


Common footprints

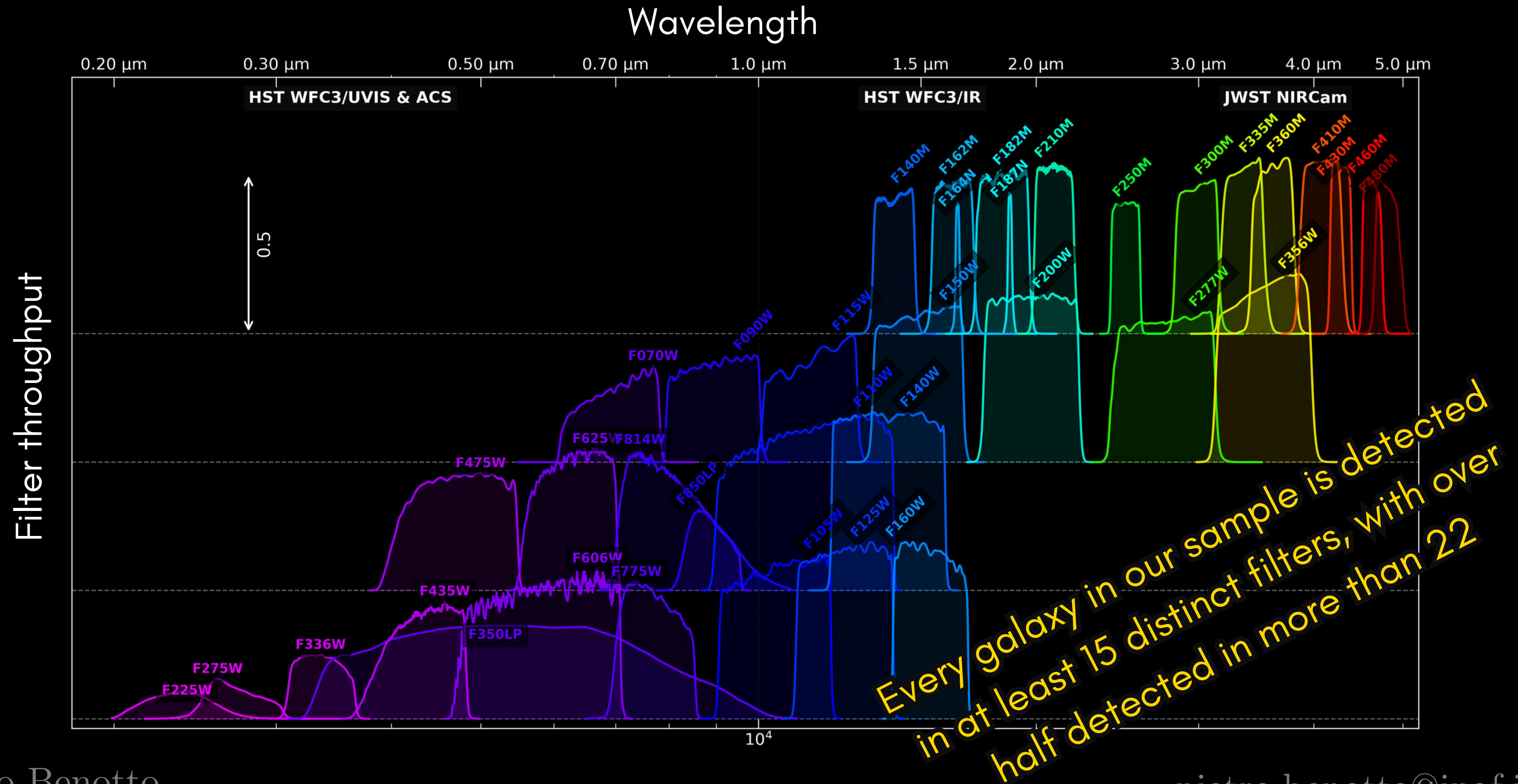
- | |
|--|
| F410M & F300M & F460M |
| F410M & F335M & F460M |
| F410M & F335M & F480M |
| F430M & F335M & F480M |
| F460M & F360M & F480M |

Redshift

Abell 2744

A2744**MACS0416****A370****GOODS-S****COSMOS****UDS****Filter footprints****Common footprints**

PSF matching (EPIMETHEUS) & SED fitting

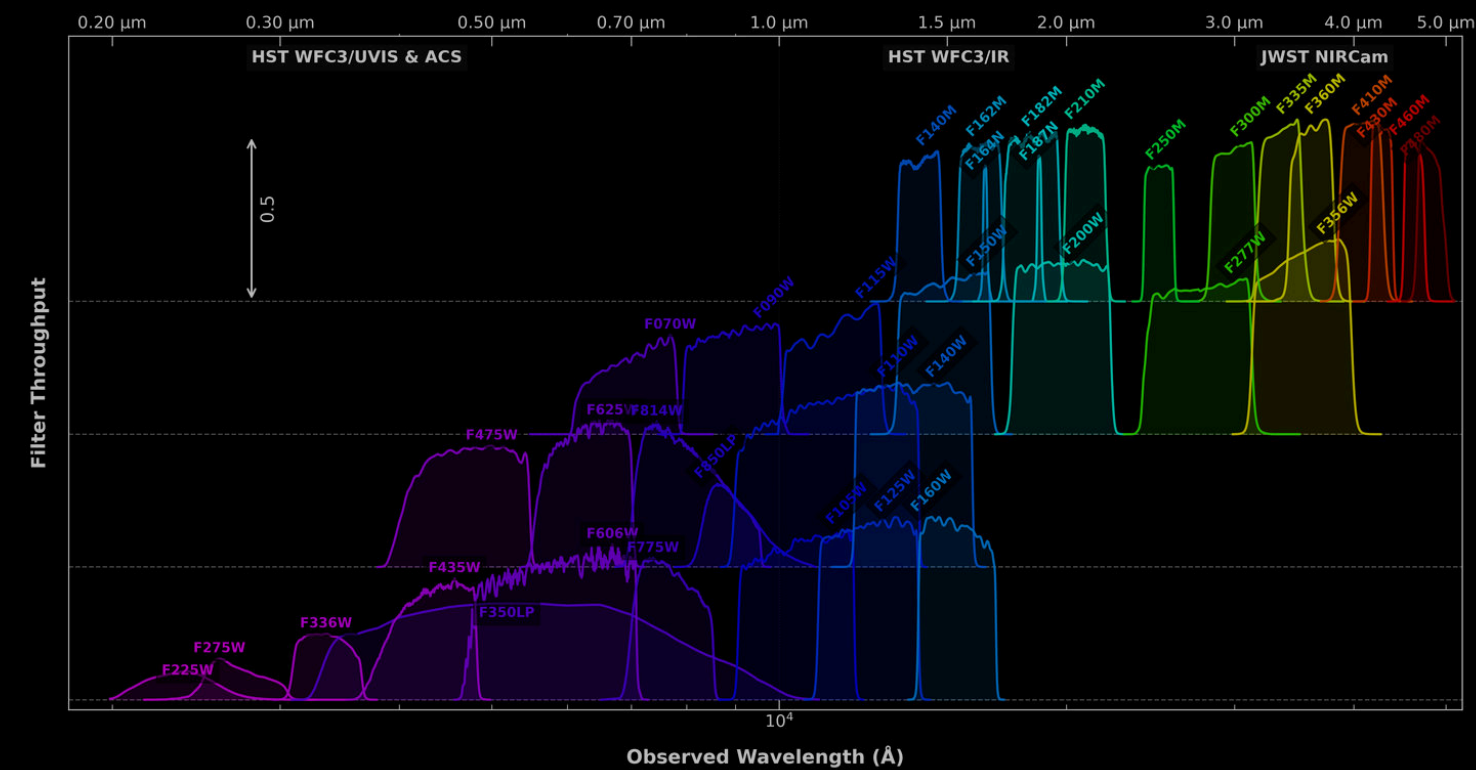


PSF matching (EPIMETHEUS) & SED fitting



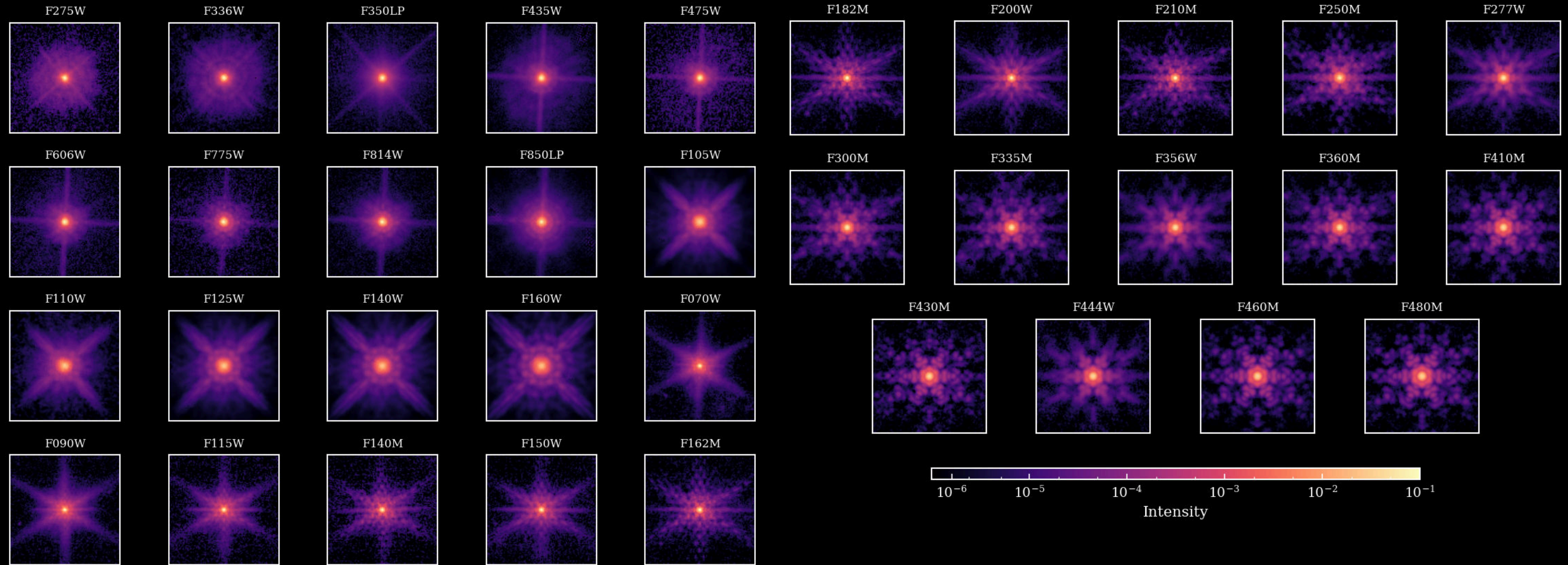
Effective
PSF
Inferred from
Mosaics using an
Estimation
Technique for
Heterogeneous
Exposures to get a
Unique PSF across the
Sensor field

- Semi-automatic star selection
- Automatic rotation computation
- Can use multiple mosaics at once
(useful in fields with few stars)



Benotto+2026b
(submitted to A&A)

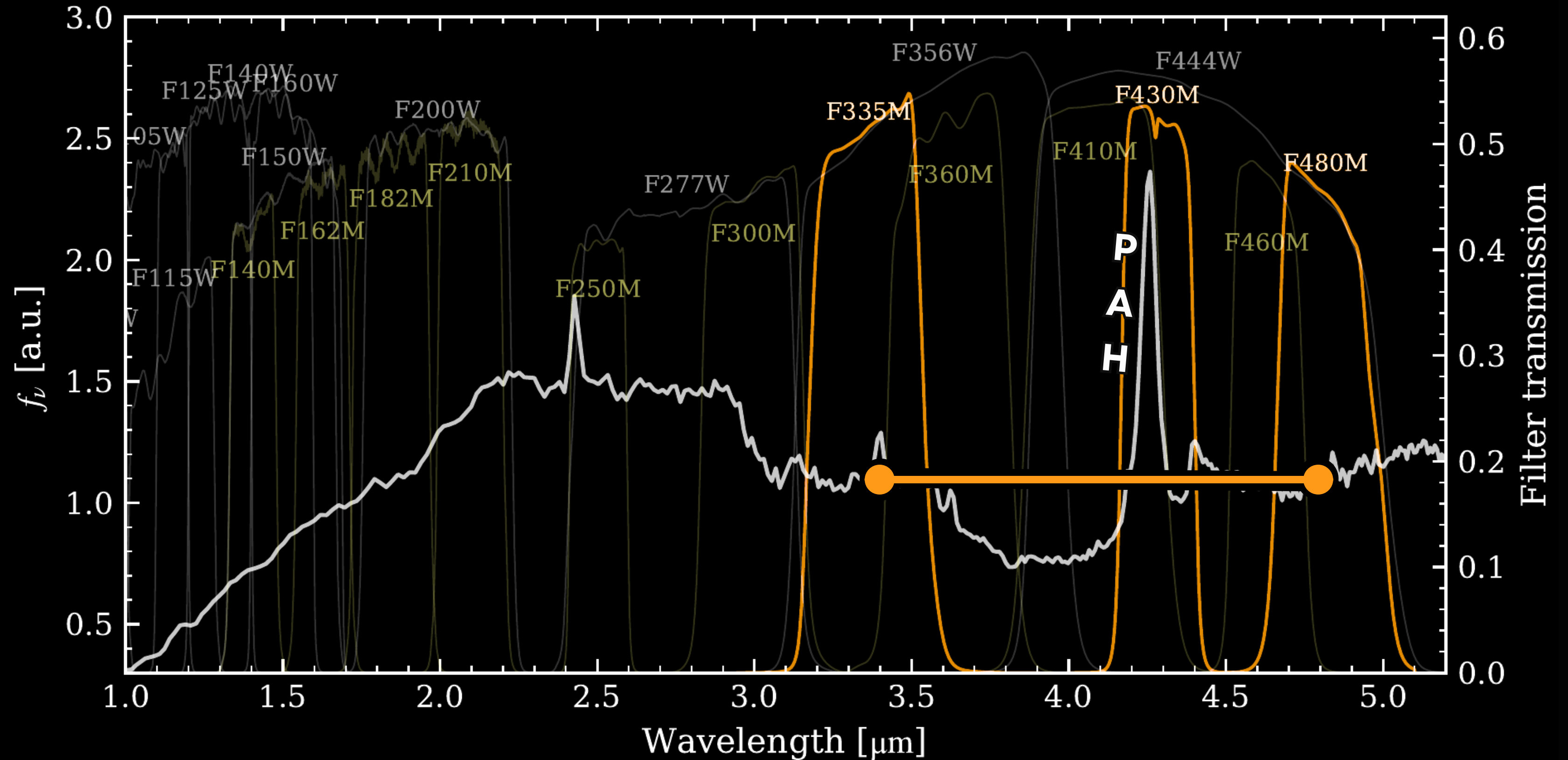
PSF matching (**EPIMETHEUS**) & SED fitting



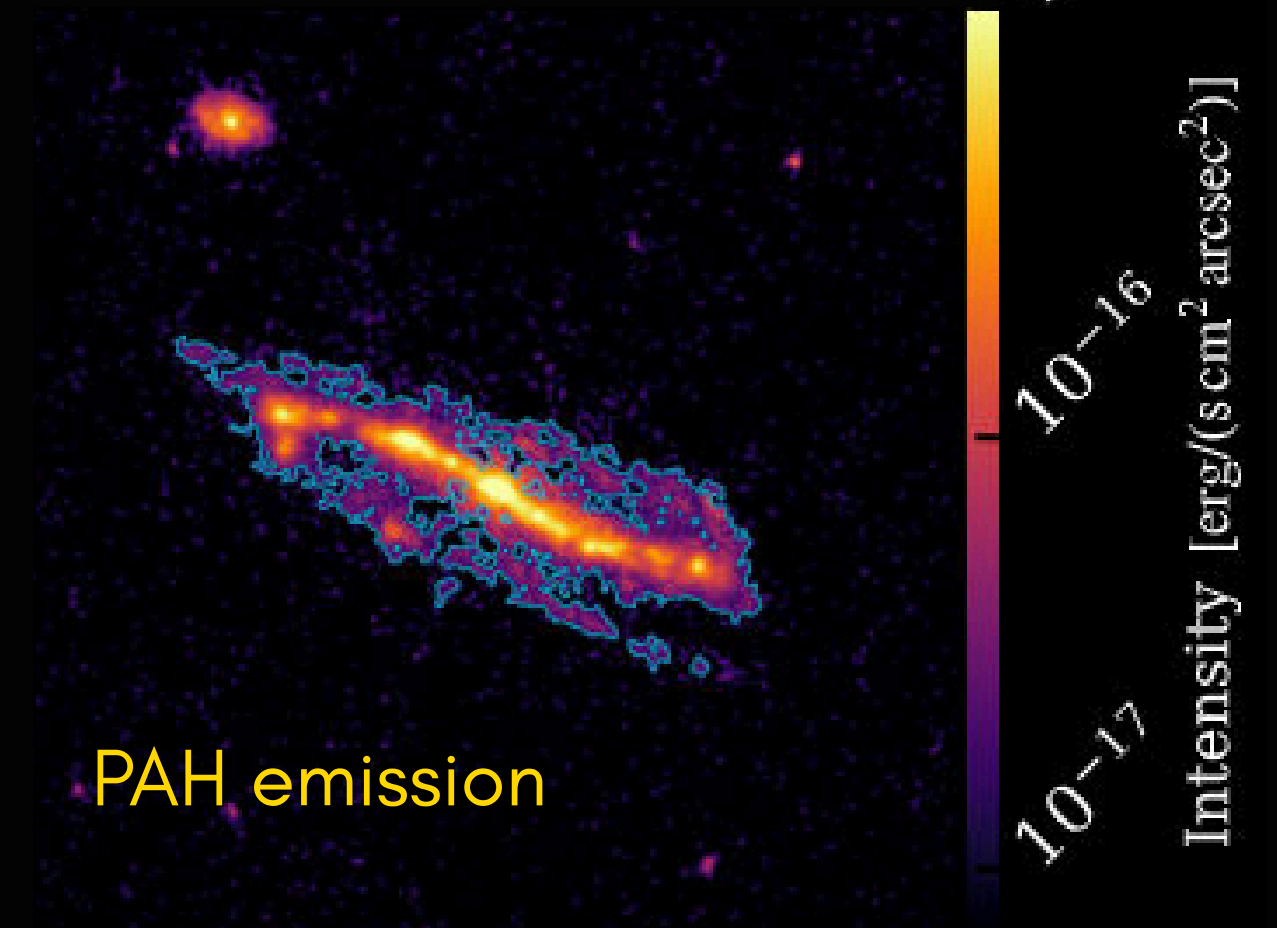
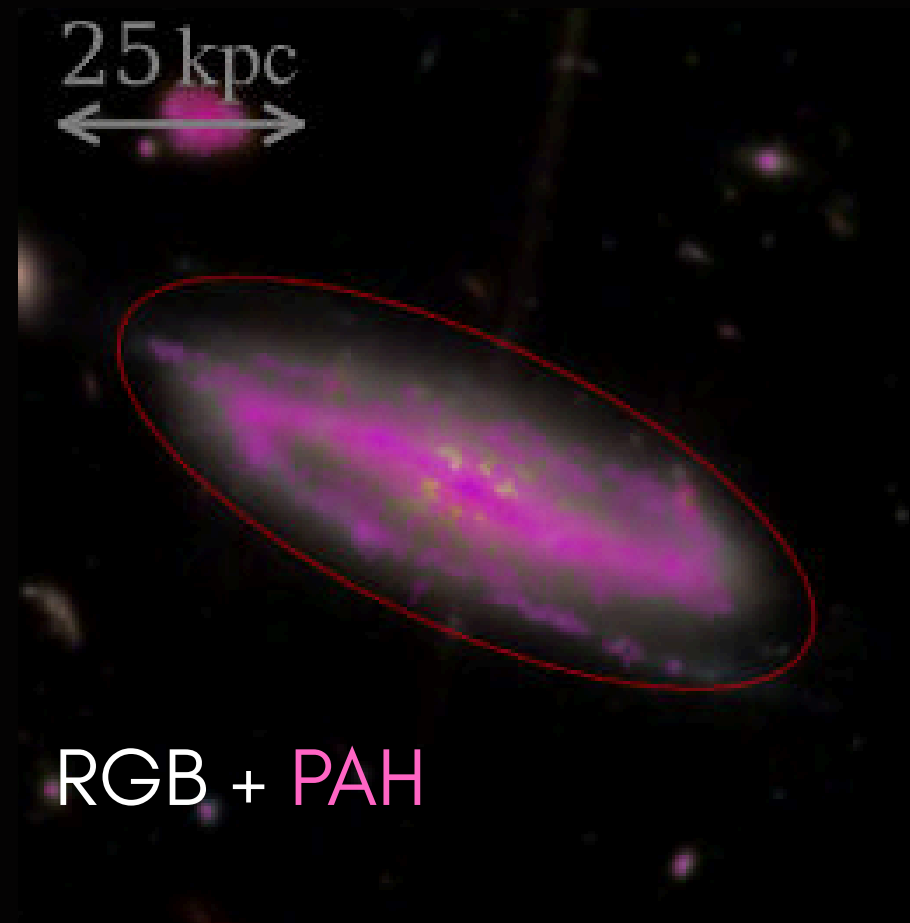
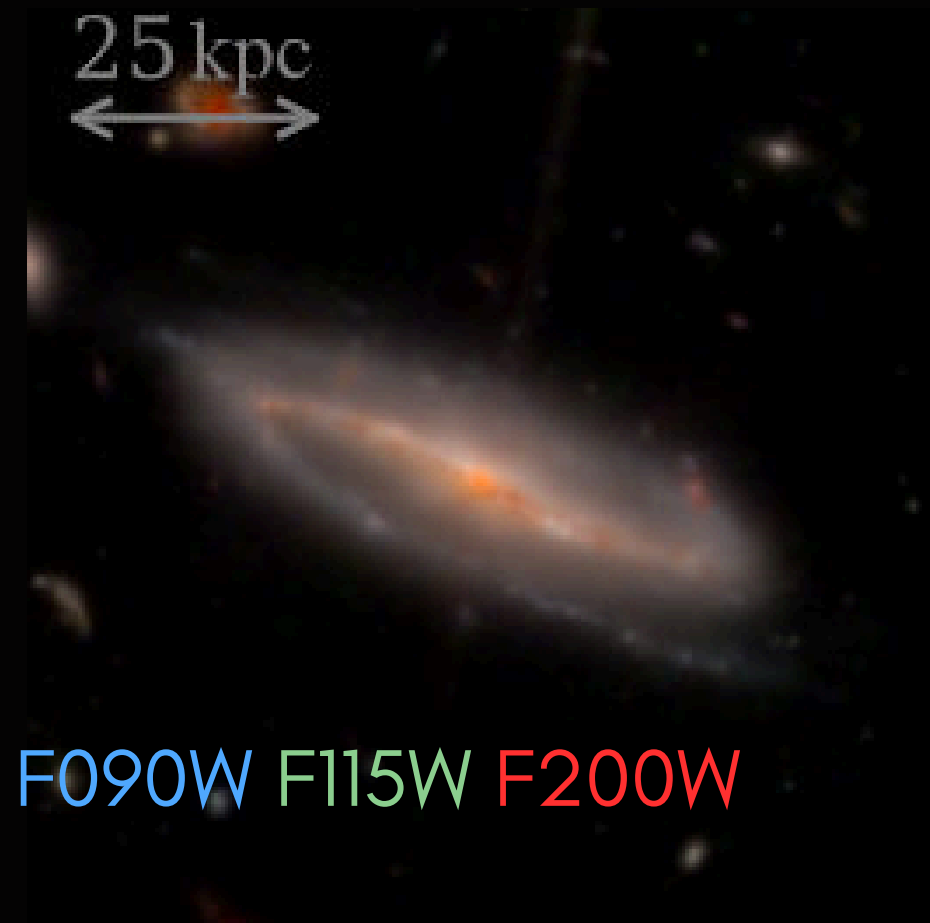
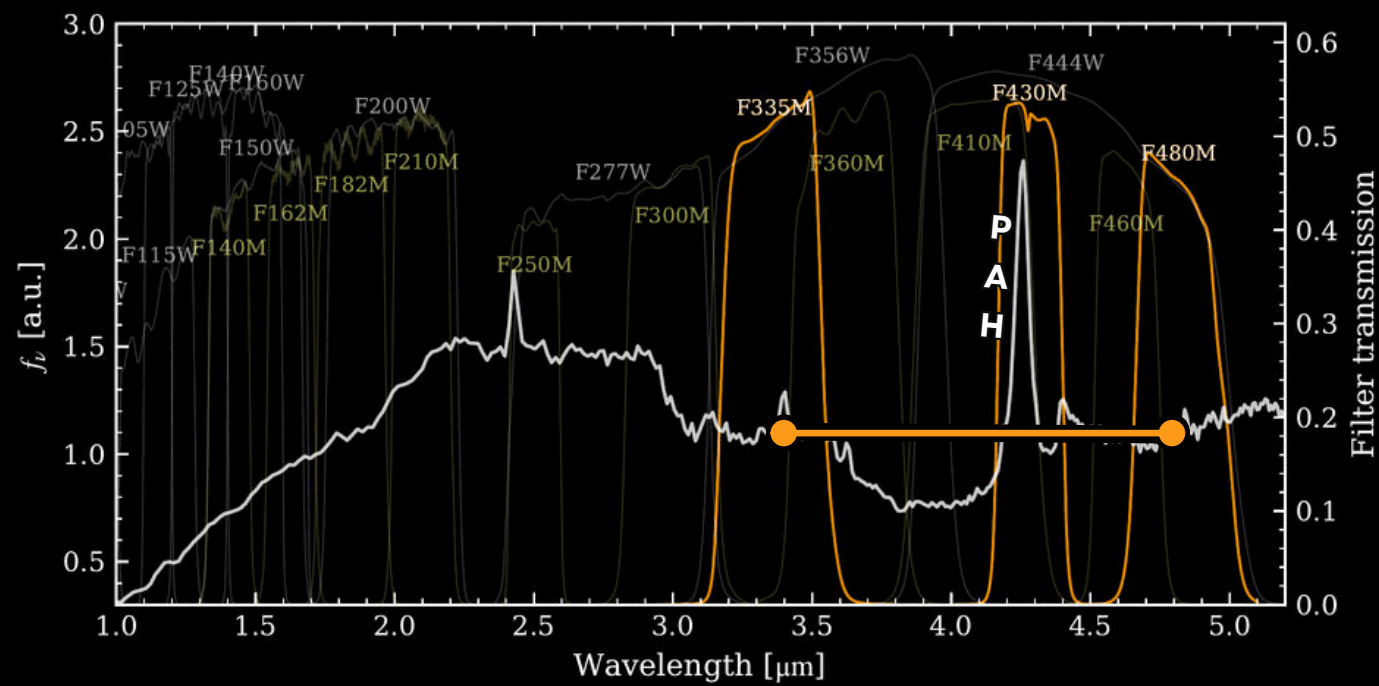
Benotto+2026b

(submitted to A&A)

PAH flux extraction



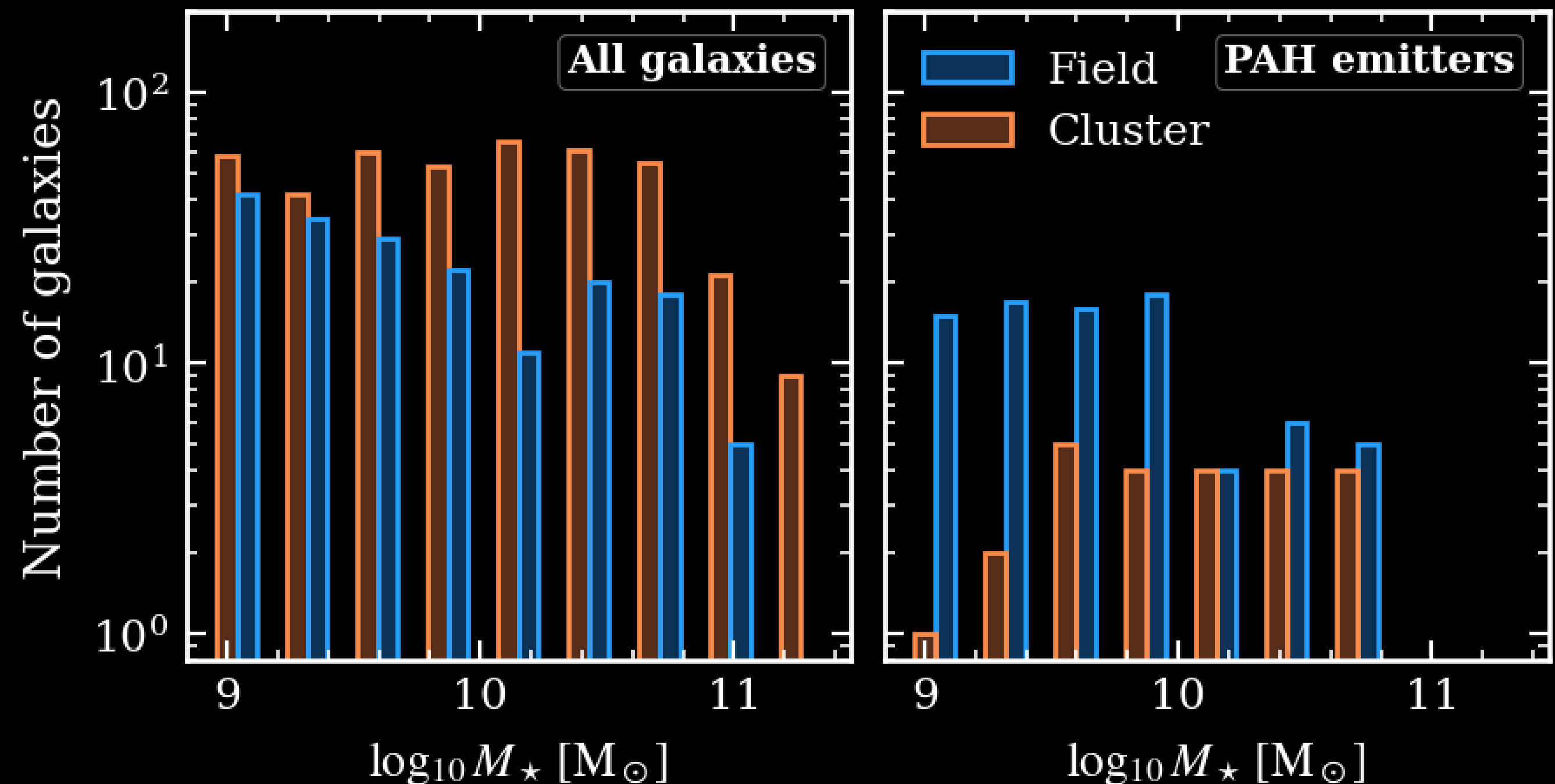
PAH flux extraction



606
galaxies

425 cluster members
181 field galaxies

Sample complete
for
 $M_{\star} > 10^{8.9} M_{\odot}$



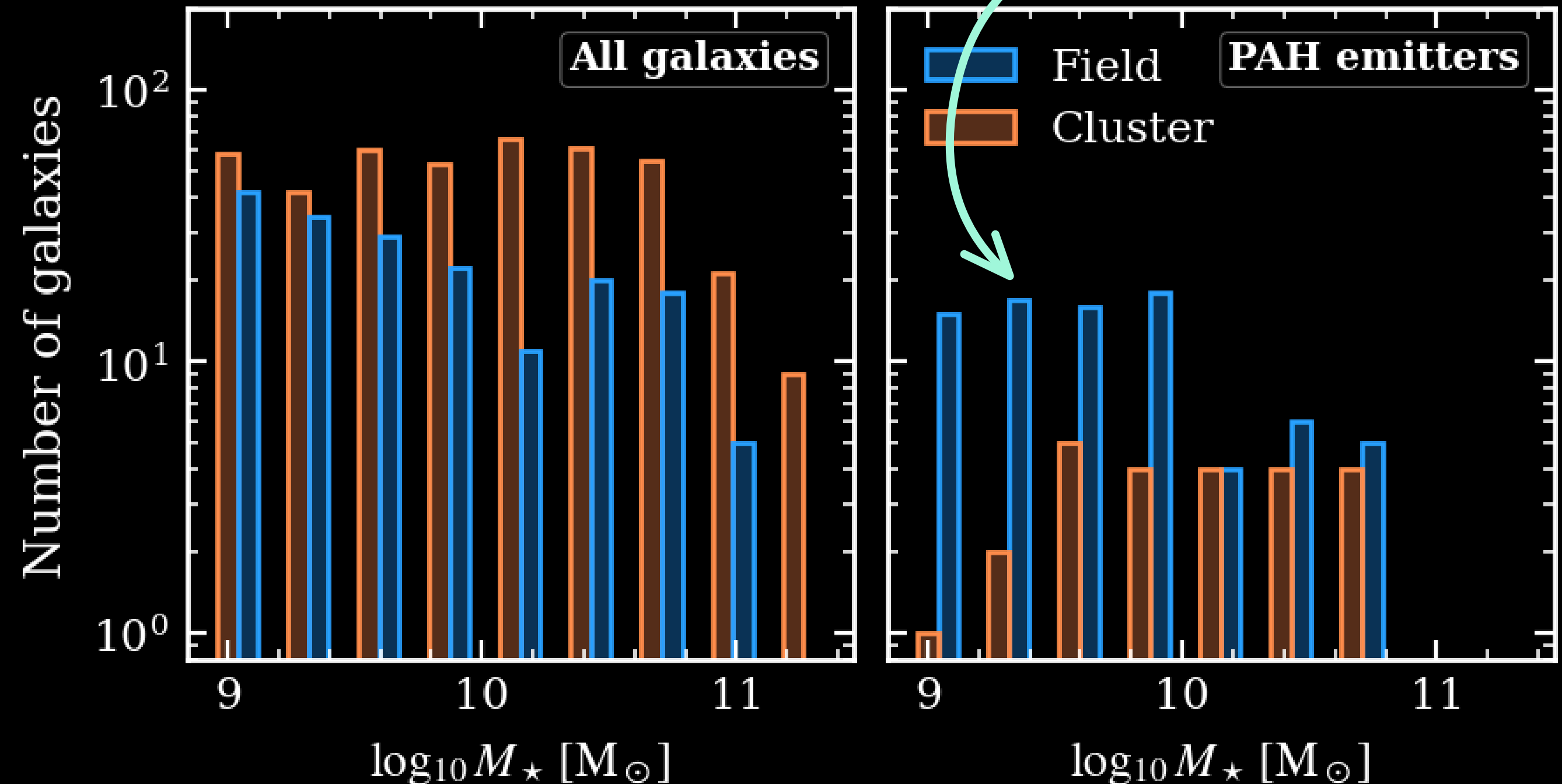
606
galaxies

425 cluster members

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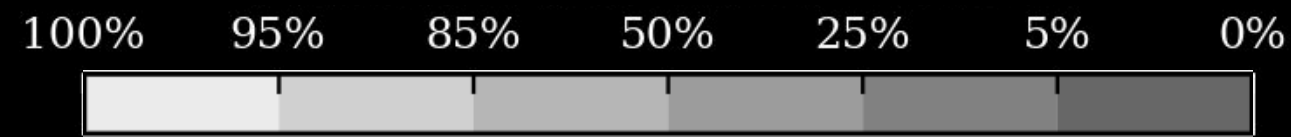
Depletion of low
mass emitters in
clusters

Sample complete
for
 $M_{\star} > 10^{8.9} M_{\odot}$

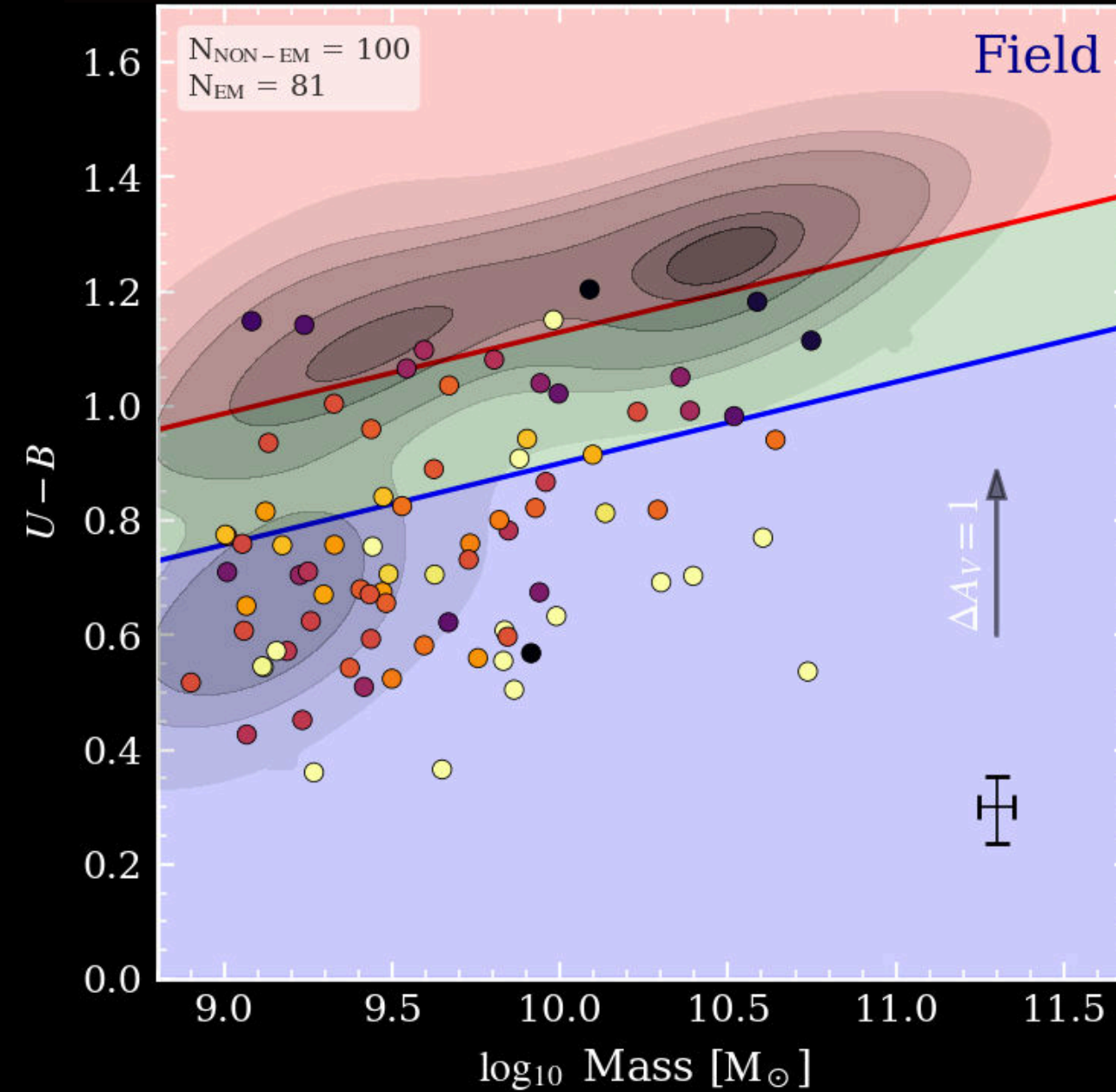
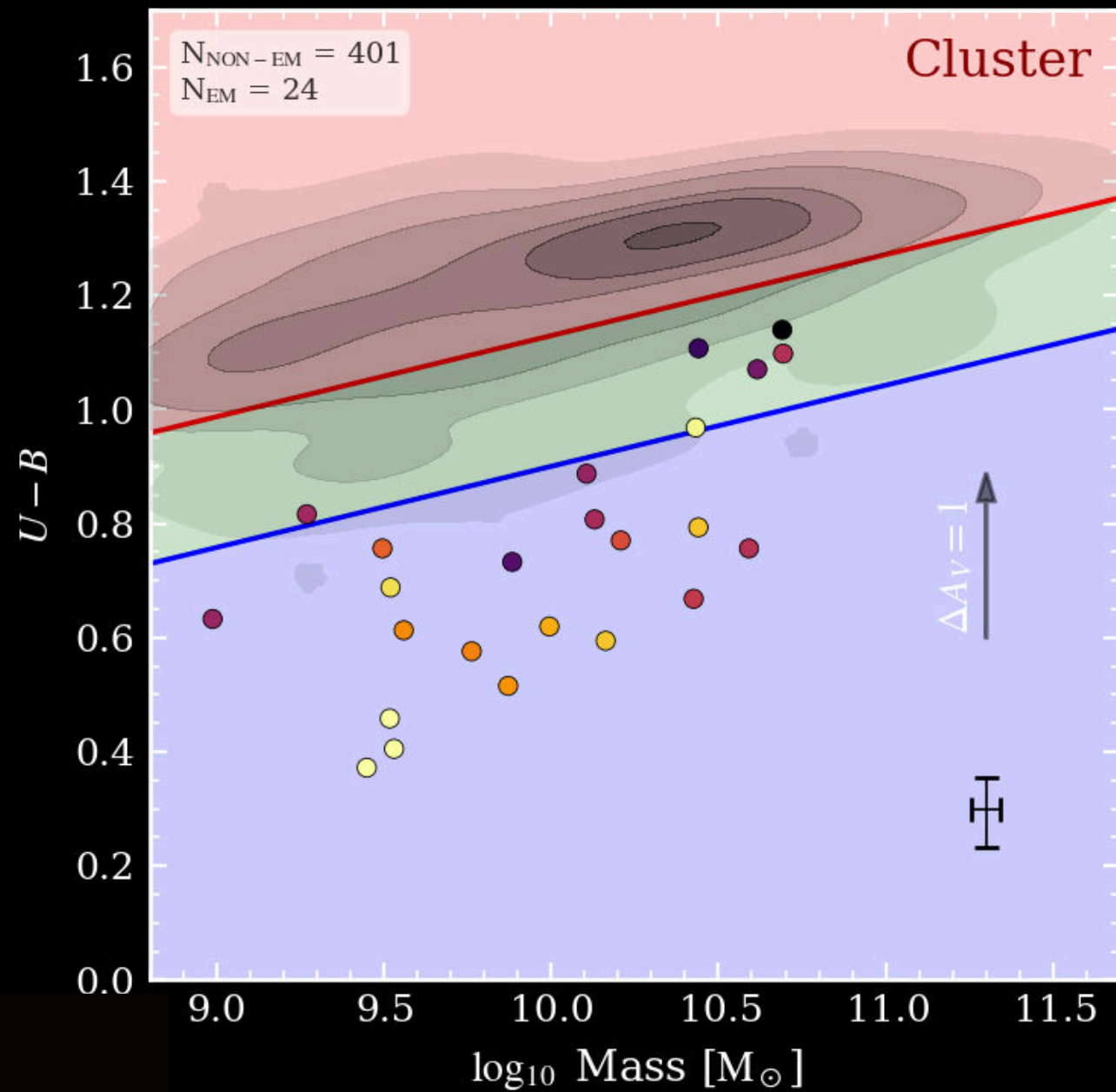
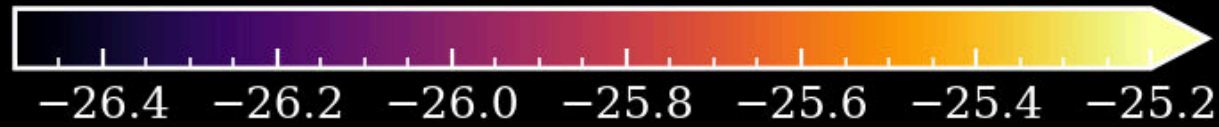


U-B vs Stellar Mass diagram

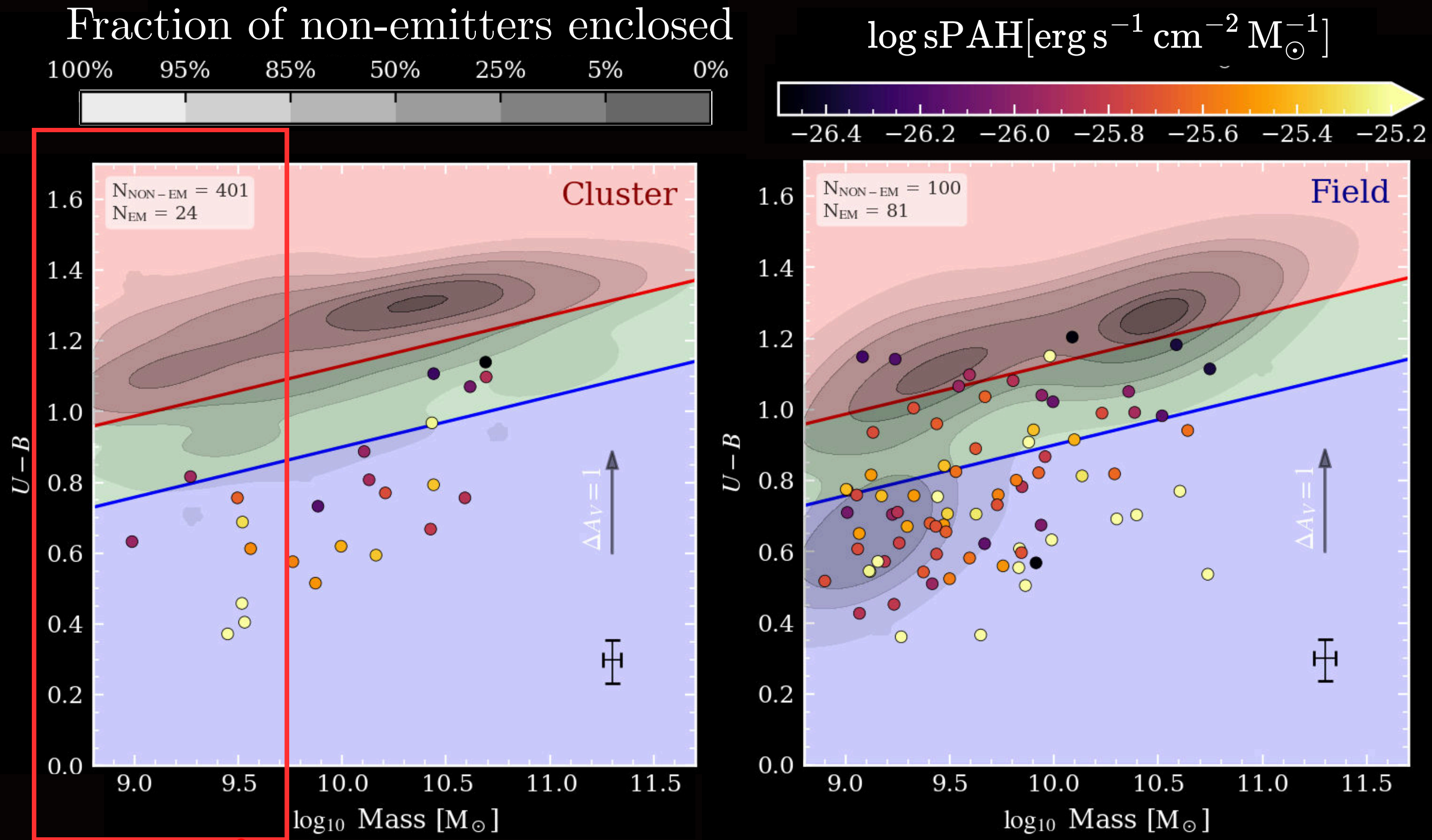
Fraction of non-emitters enclosed



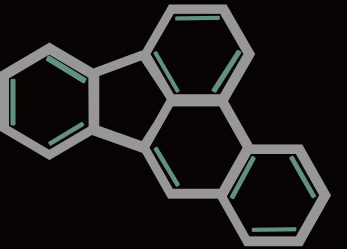
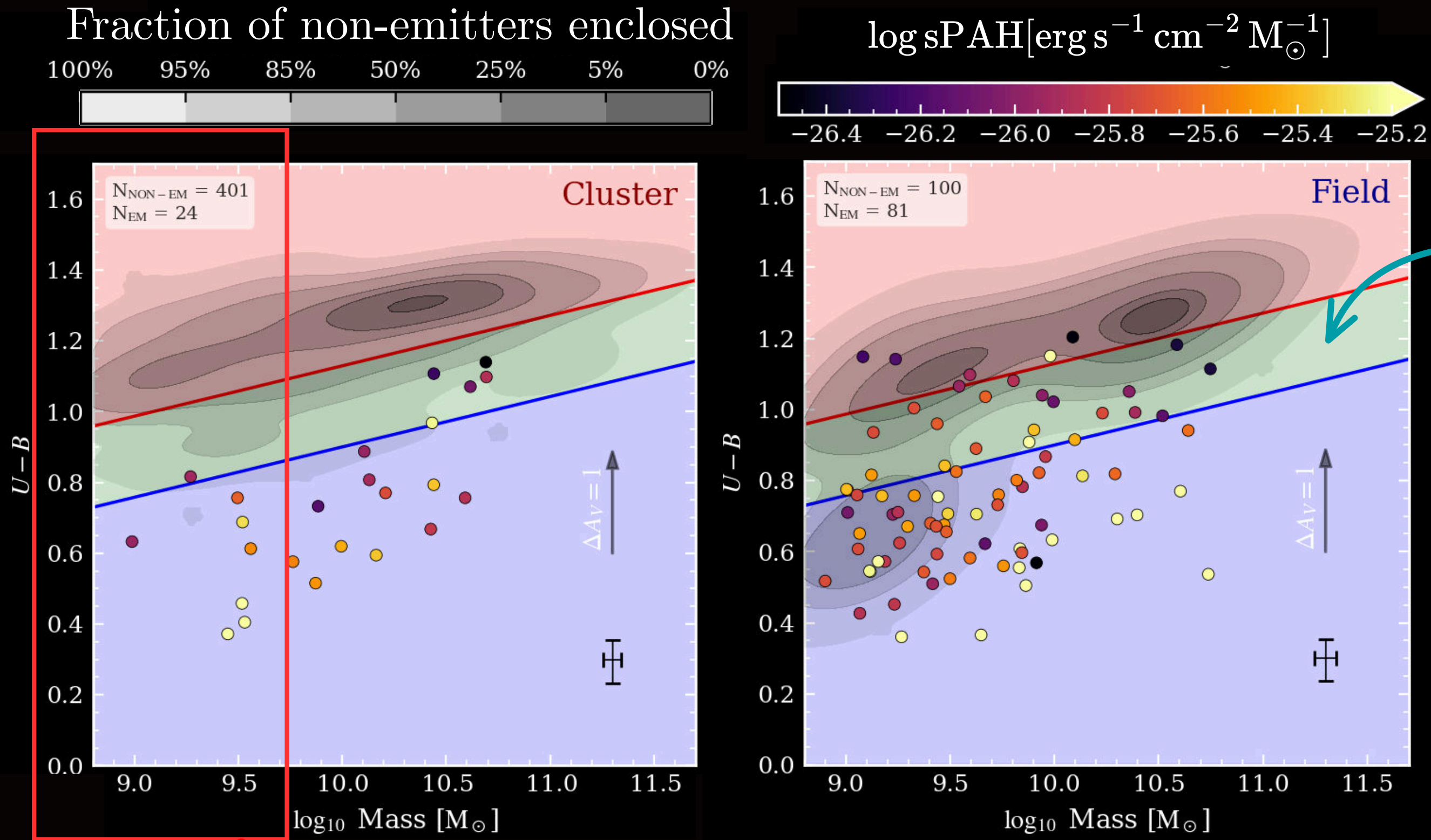
$\log \text{sPAH} [\text{erg s}^{-1} \text{cm}^{-2} \text{M}_{\odot}^{-1}]$



U-B vs Stellar Mass diagram



U-B vs Stellar Mass diagram



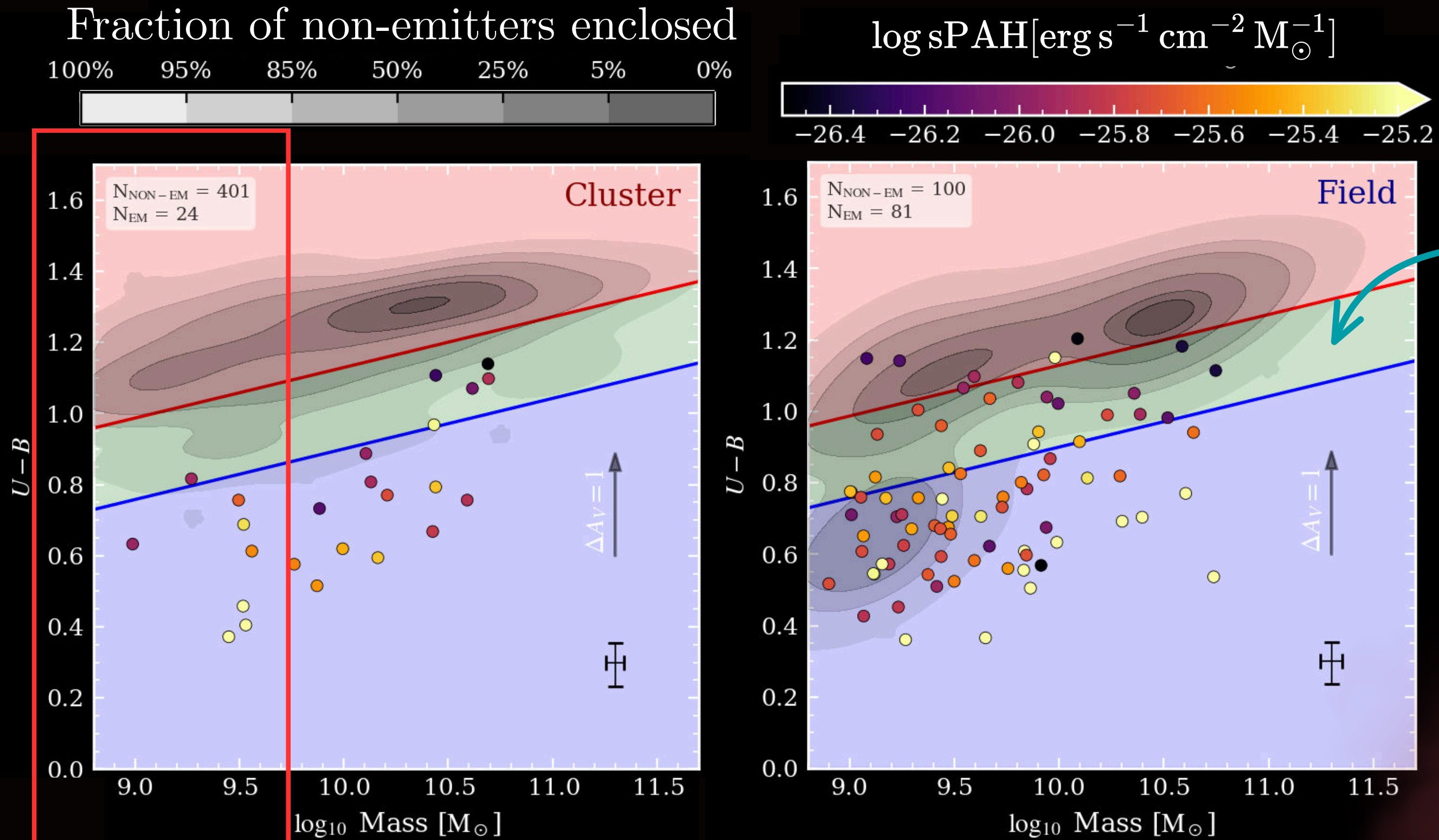
Fraction of PAH emitters
in the green zone:
(47±6)% field vs (6±3)%
in the cluster

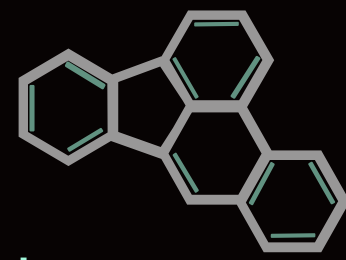
Pietro Benotto

Low-mass cluster galaxies
dominated by quenched spheroids

pietro.benotto@inaf.it

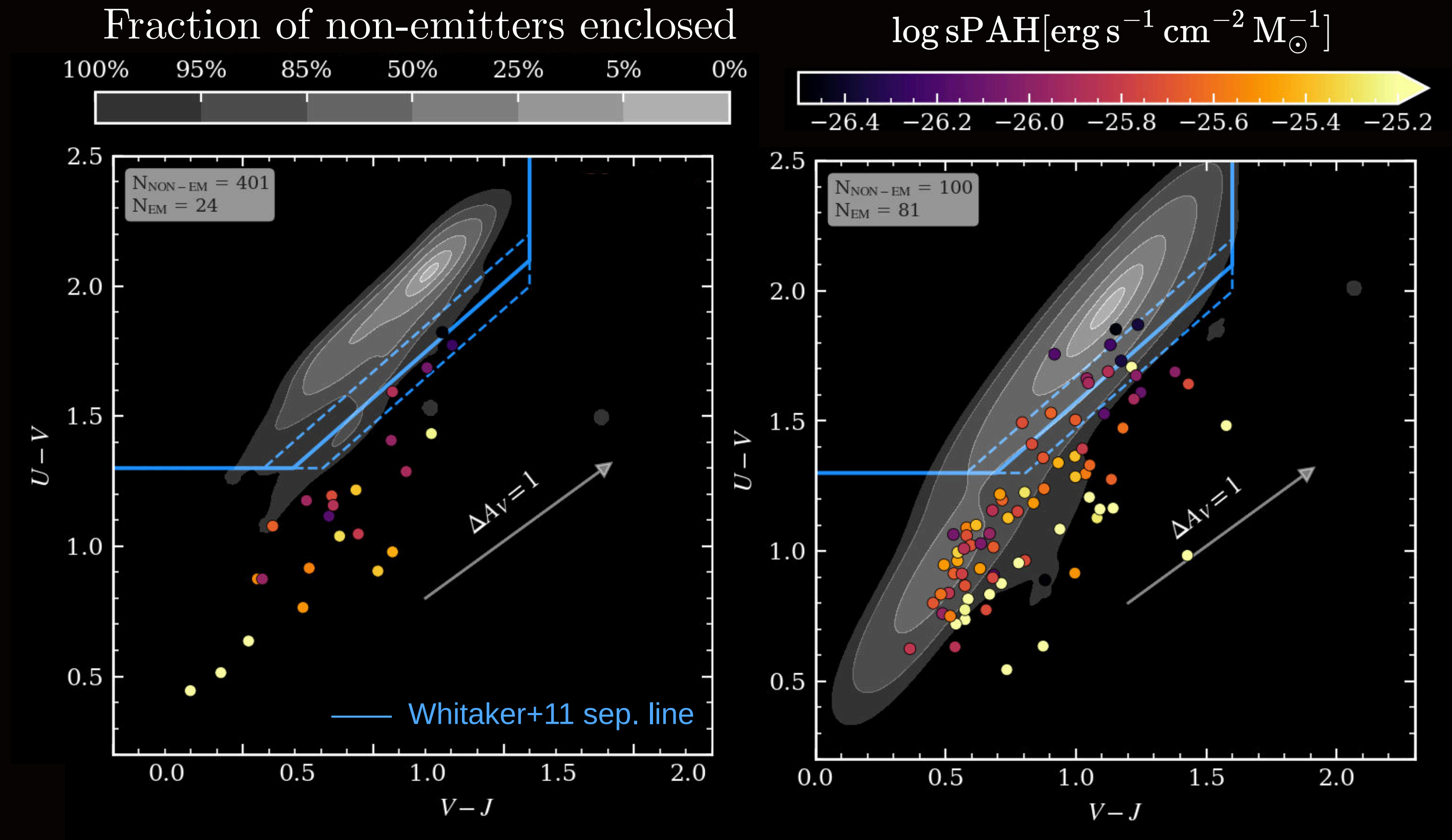
U-B vs Stellar Mass diagram




Fraction of PAH emitters
in the green zone:
(47±6)% field vs (6±3)%
in the cluster

**Dust attenuation may
affects the result!**

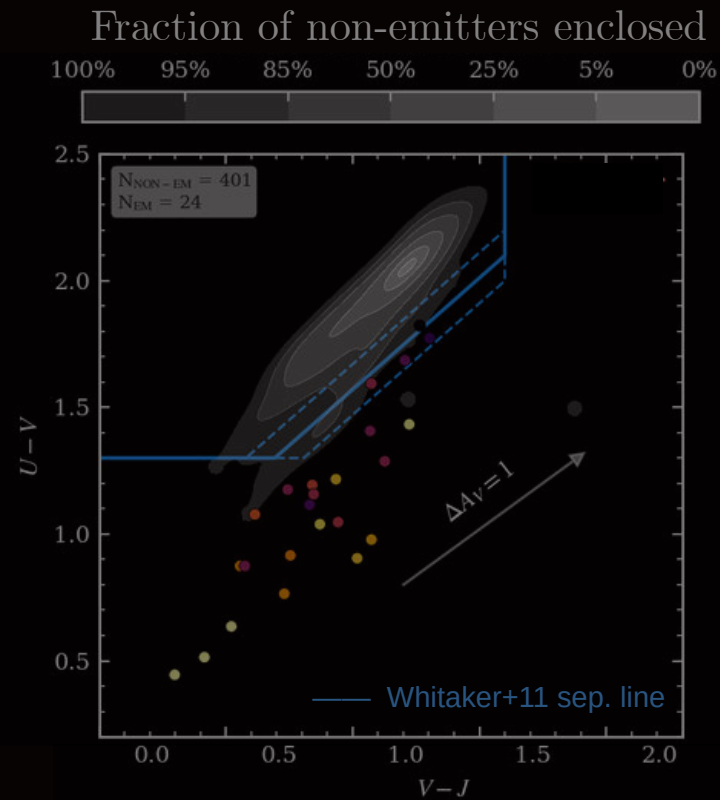
UVJ diagram



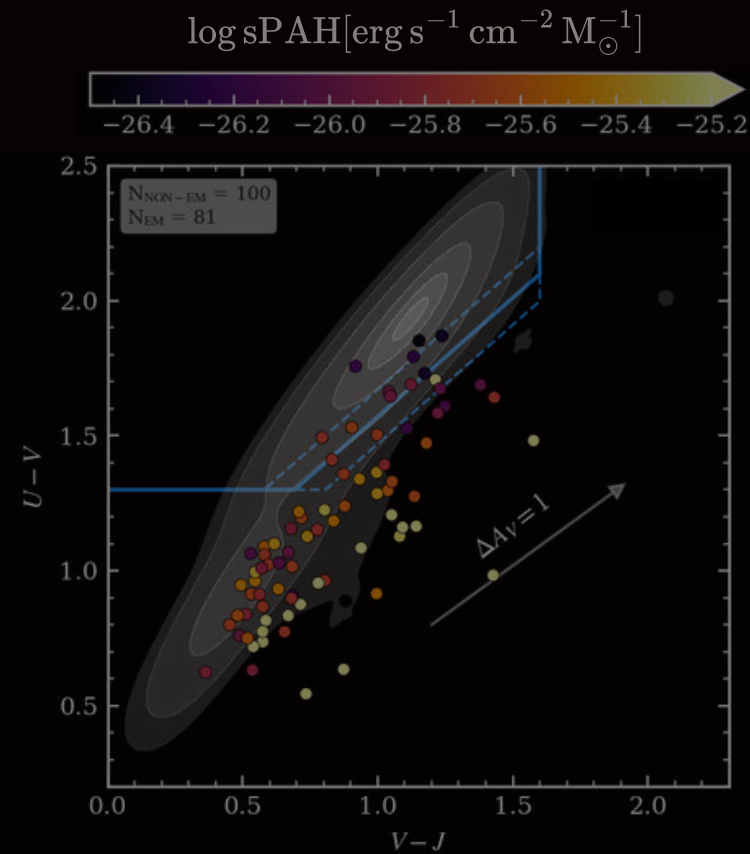
CLUSTERS GALAXIES

FIELD GALAXIES

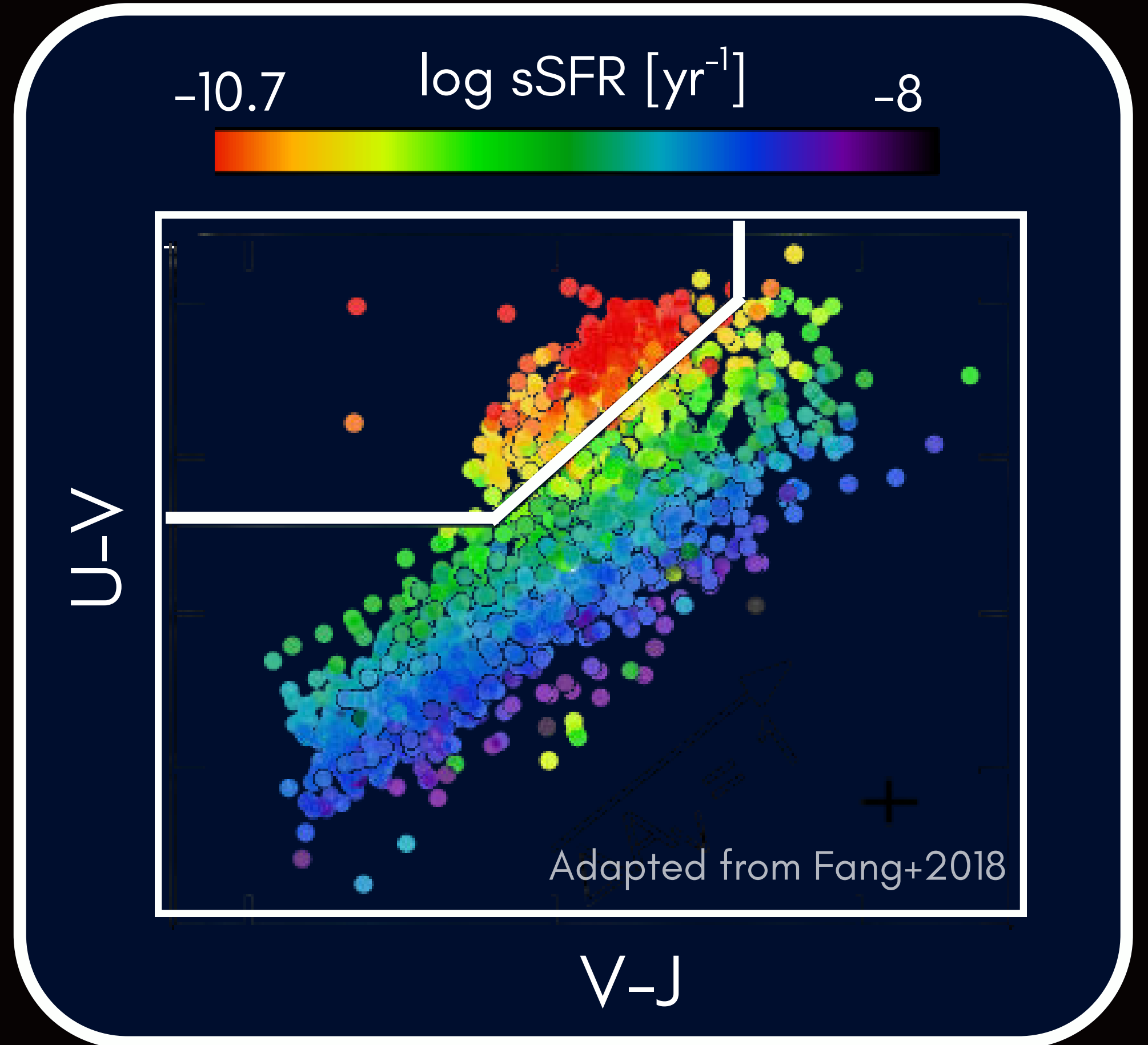
UVJ diagram



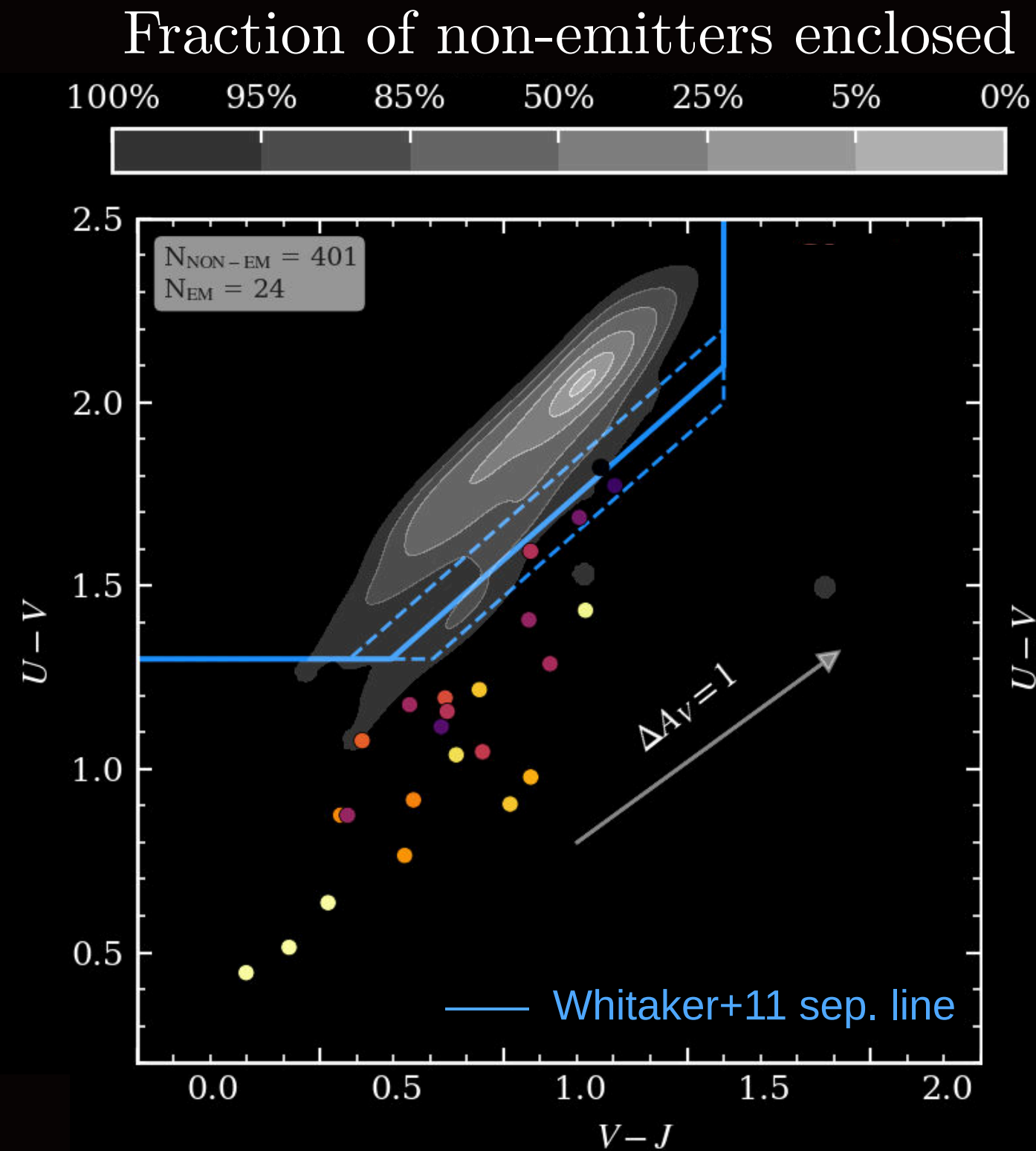
CLUSTERS GALAXIES



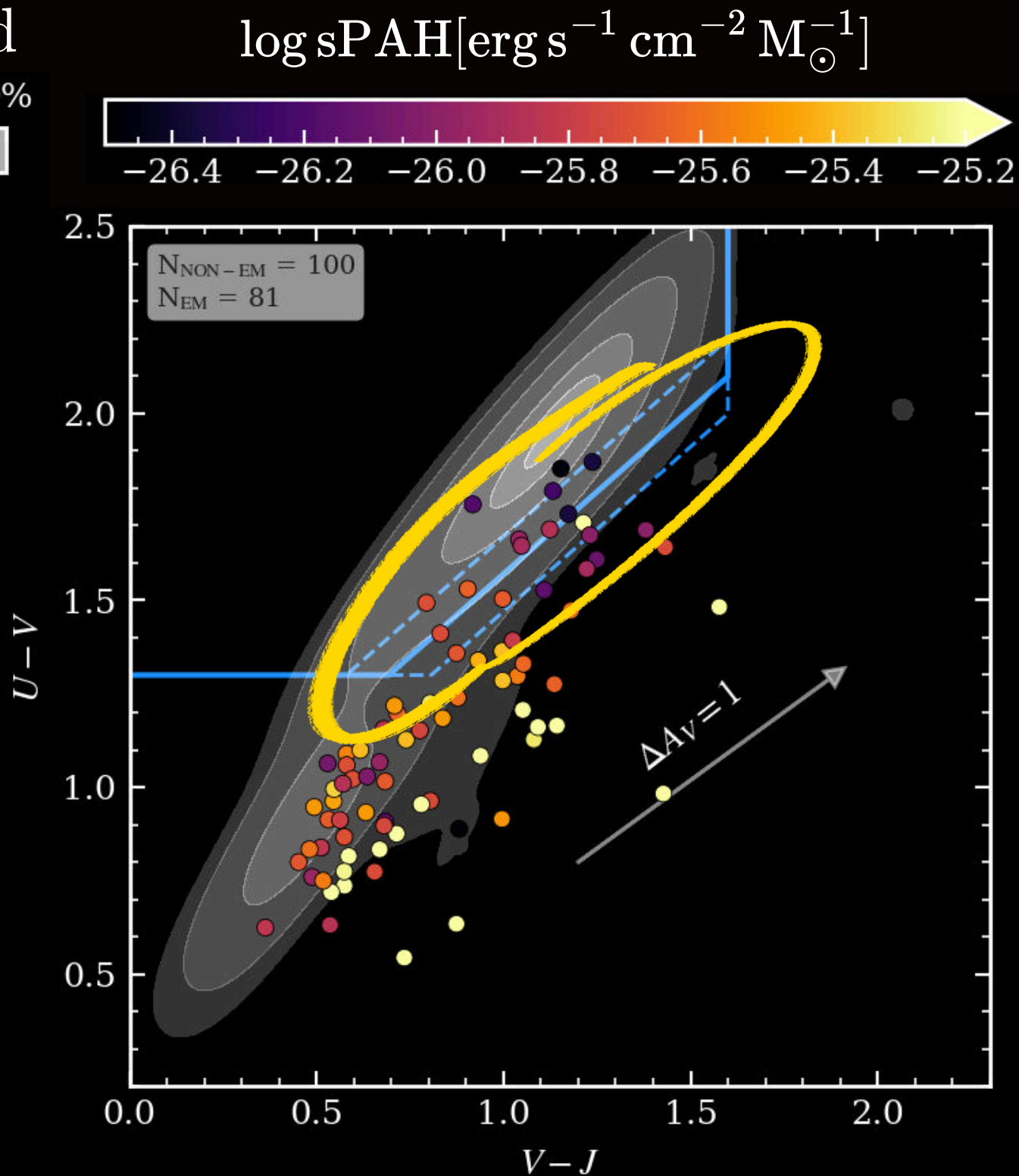
FIELD GALAXIES



UVJ diagram



CLUSTERS GALAXIES

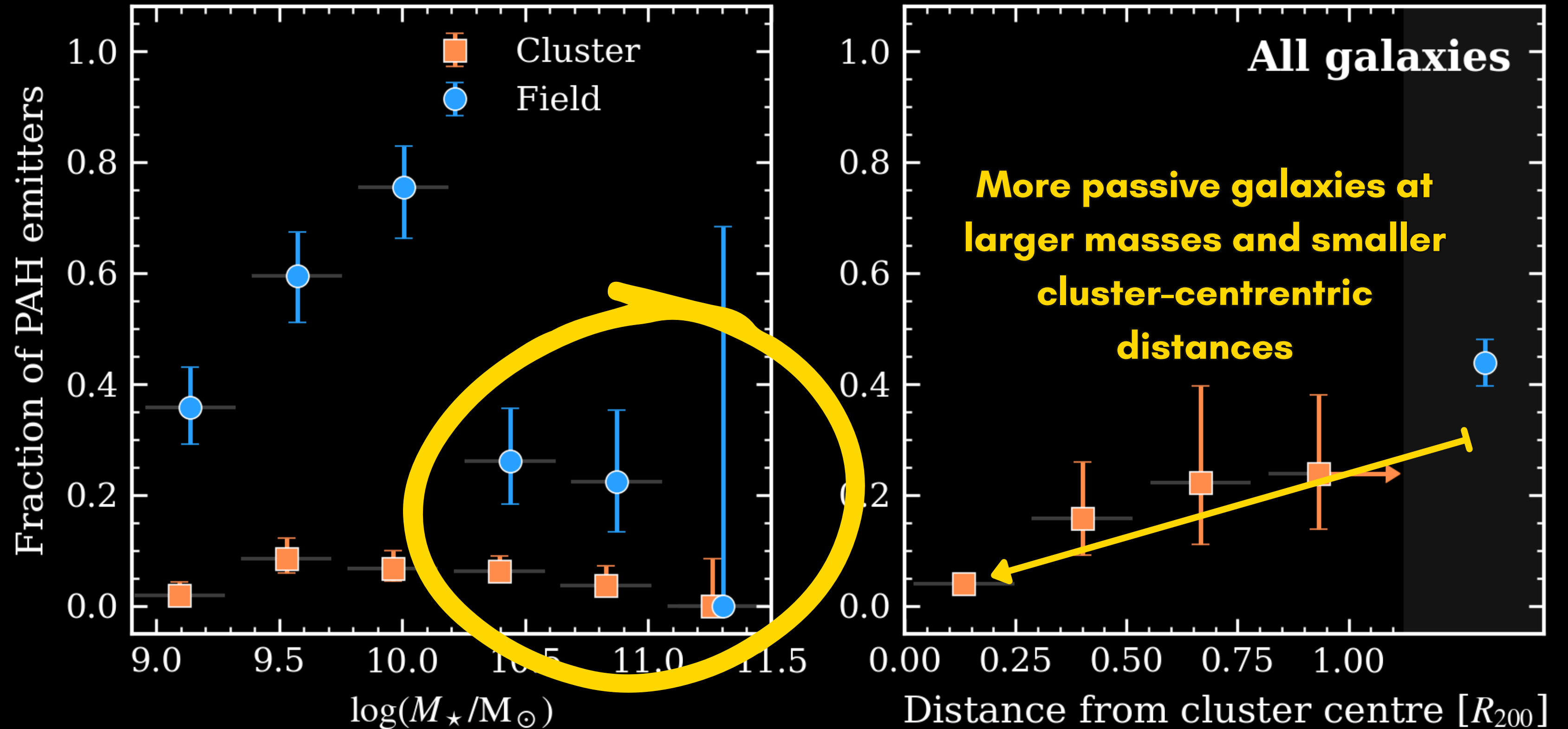


FIELD GALAXIES

Transition zone similar to
McNab+22 and Mao+22

Defined as the region
extending within ± 0.1
magnitudes on either
side of the Whitaker+11
separation line

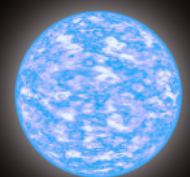
Fraction of PAH emitters



Fraction of PAH emitters

CLUSTER

FIELD



STAR-FORMING

59%

69%

($\pm 6\%$)



TRANSITION

9%

54%

($\pm 6\%$)



QUIESCENT

0%

5%

($\pm 2\%$)

	CLUSTER	FIELD	
<u>STAR-FORMING</u>	59%	69%	(±6%)
<u>TRANSITION</u>	9%	54%	(±6%)
<u>QUIESCENT</u>	0%	5%	(±2%)



$$M_{\star} < 10^{10.2} M_{\odot}$$

CLUSTER	FIELD
$0^{+3}_{-0}\%$	$72^{+9}_{-12}\%$

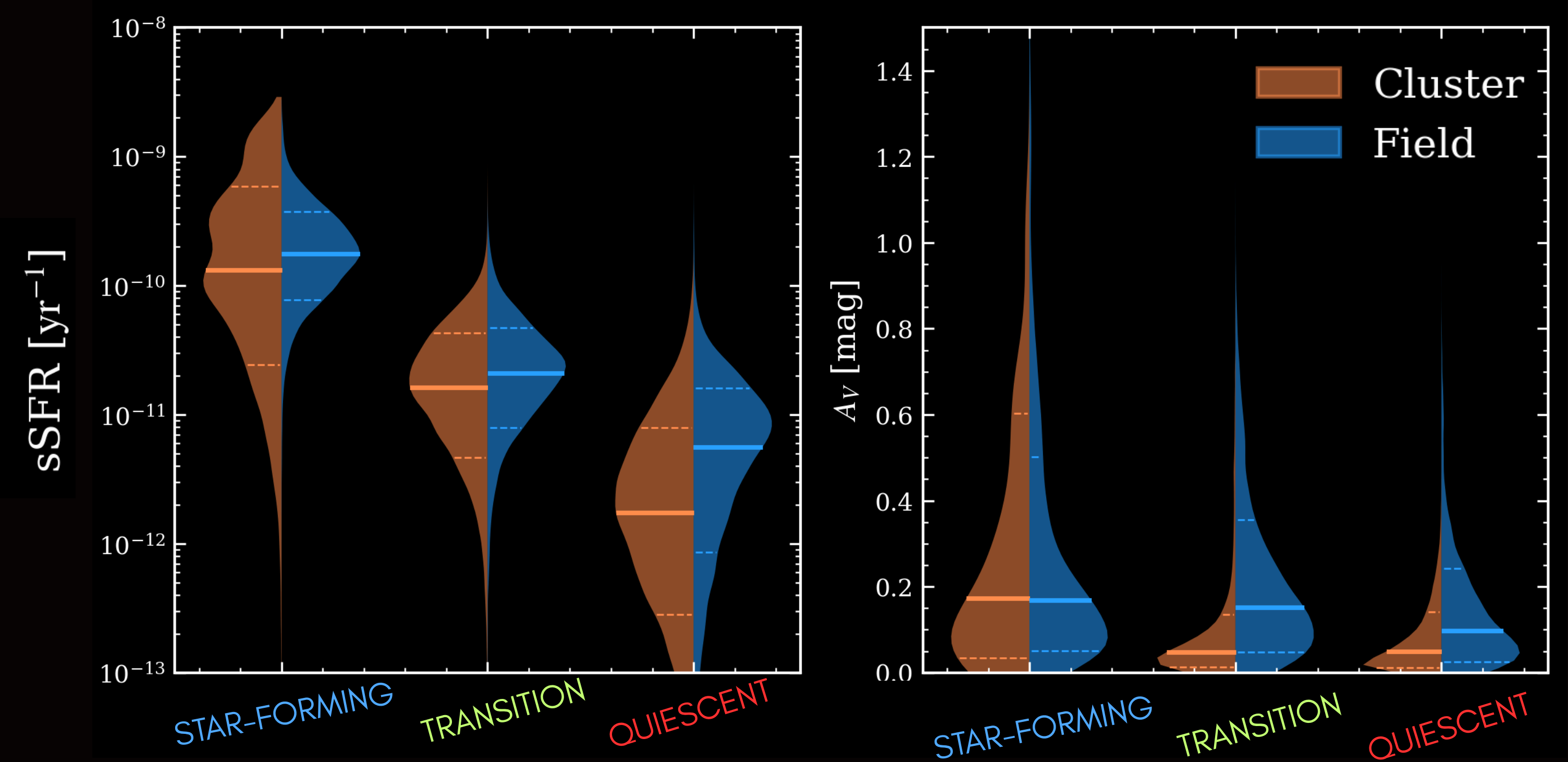
$> 5\sigma$
significance

Benotto+2026c
(in prep.)

	CLUSTER	FIELD	
<u>STAR-FORMING</u>	59%	69% ($\pm 6\%$)	
<u>TRANSITION</u>	9%	54% ($\pm 6\%$)	
<u>QUIESCENT</u>	0%	5% ($\pm 2\%$)	

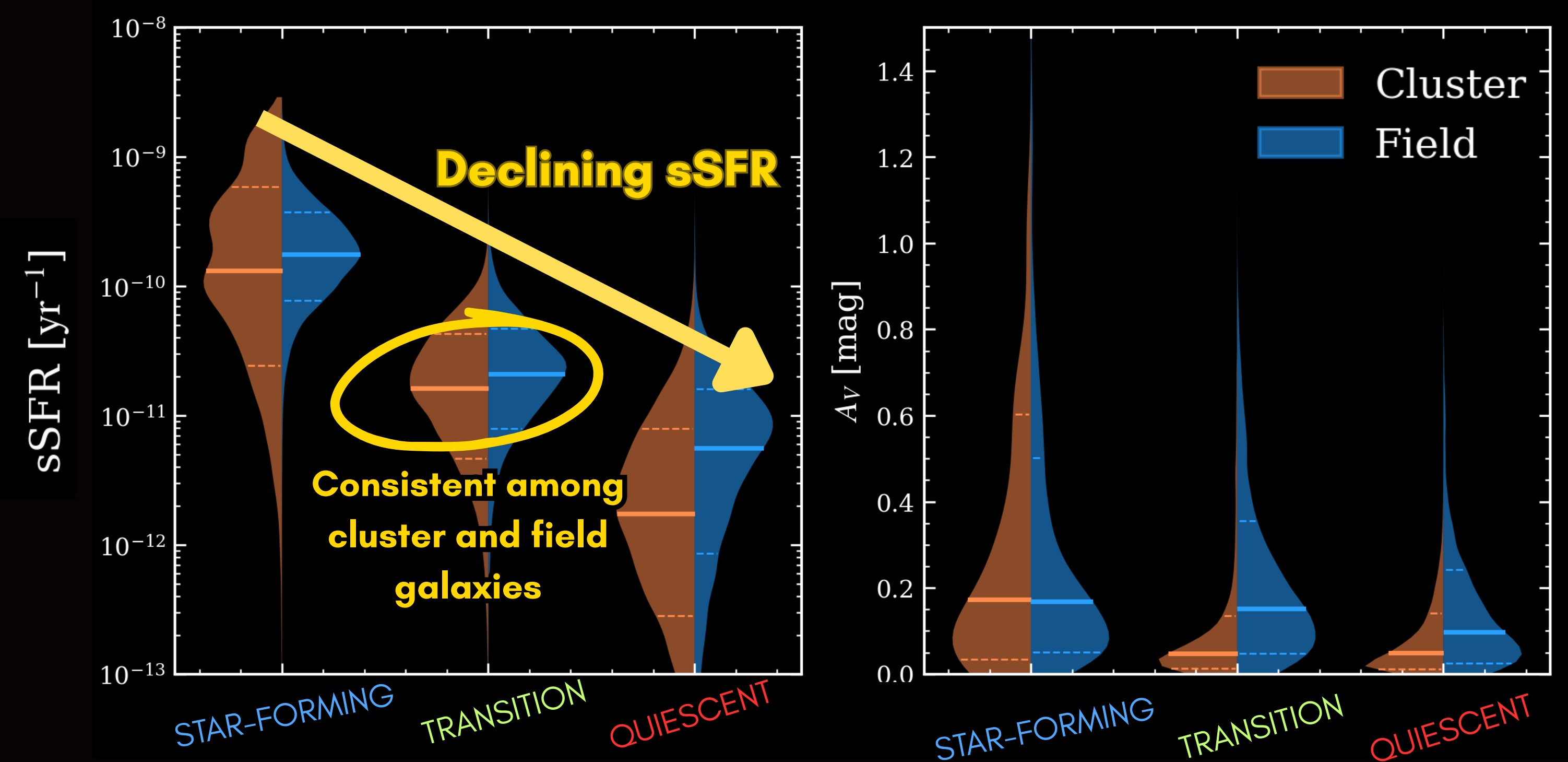
Benotto+2026c
(in prep.)

Pietro Benotto



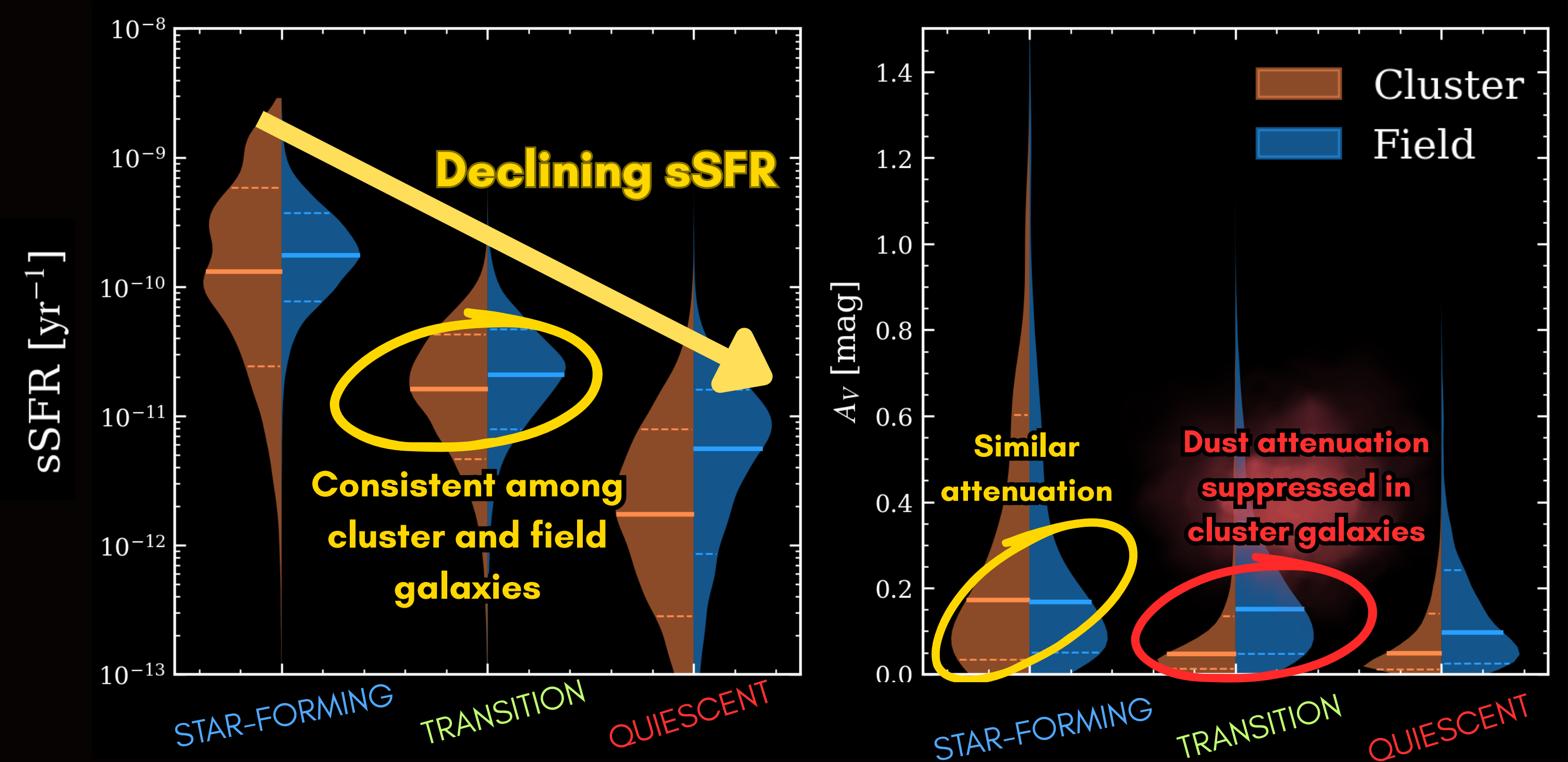
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Benotto+2026c
(in prep.)

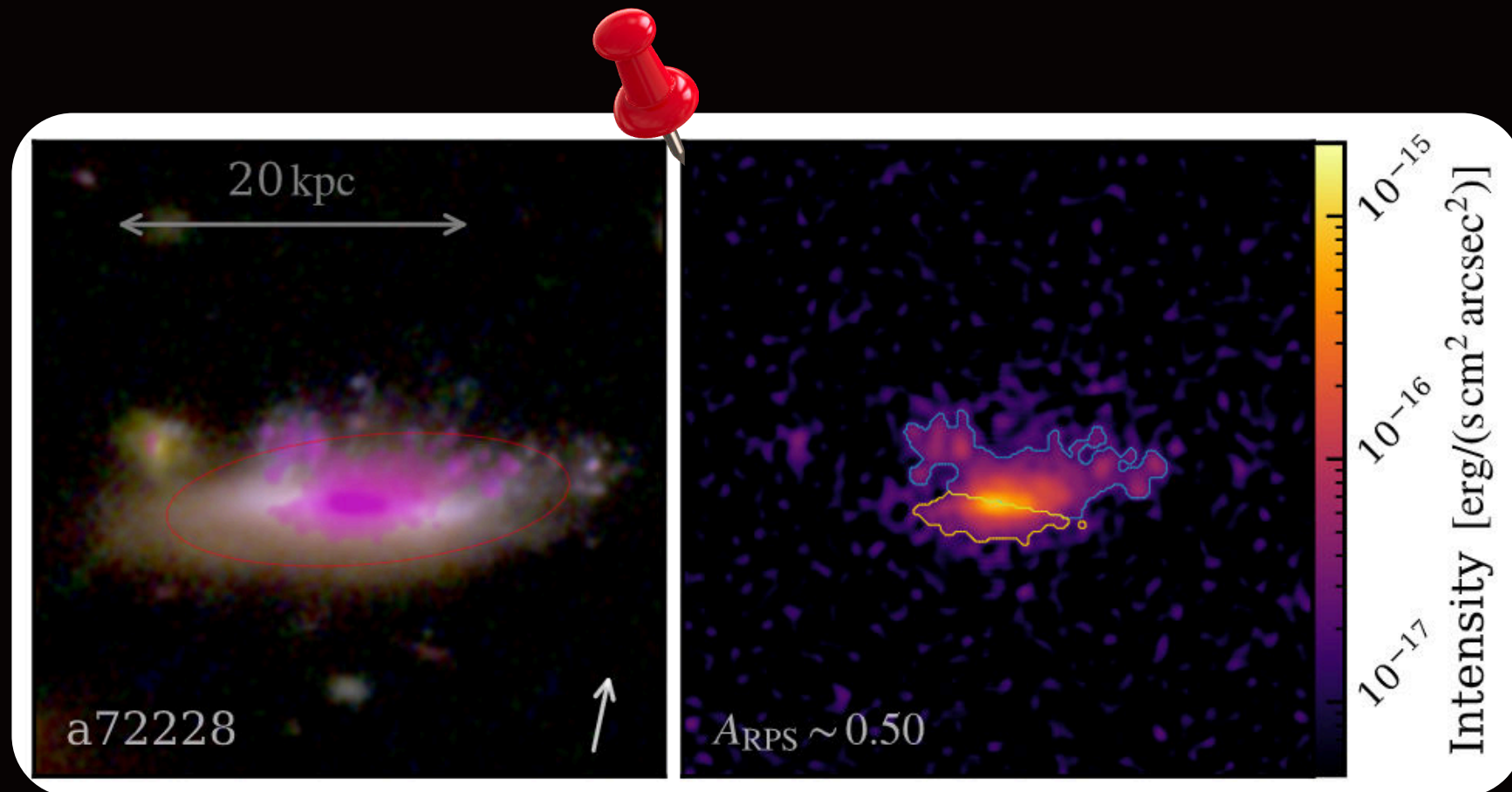
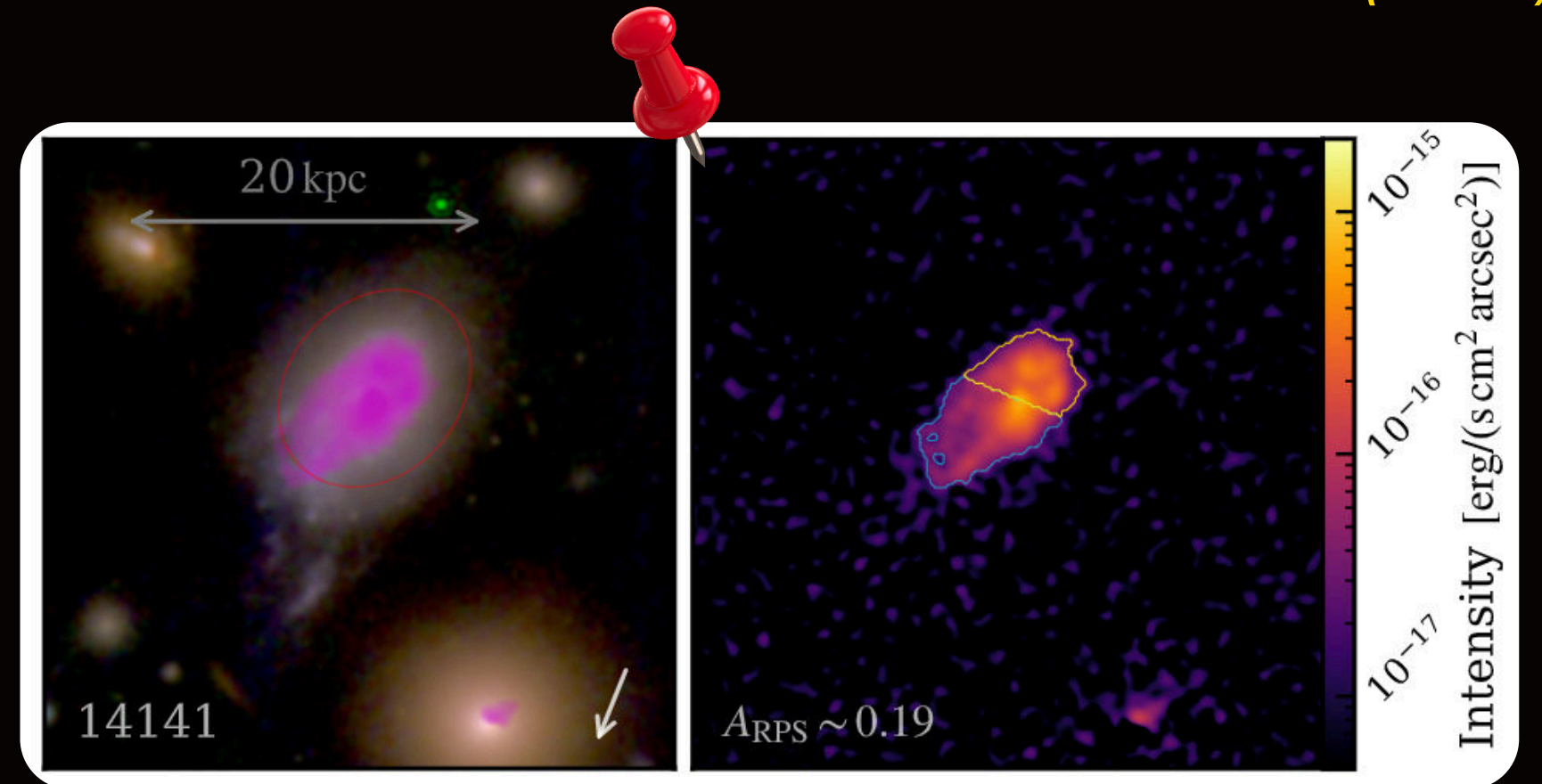
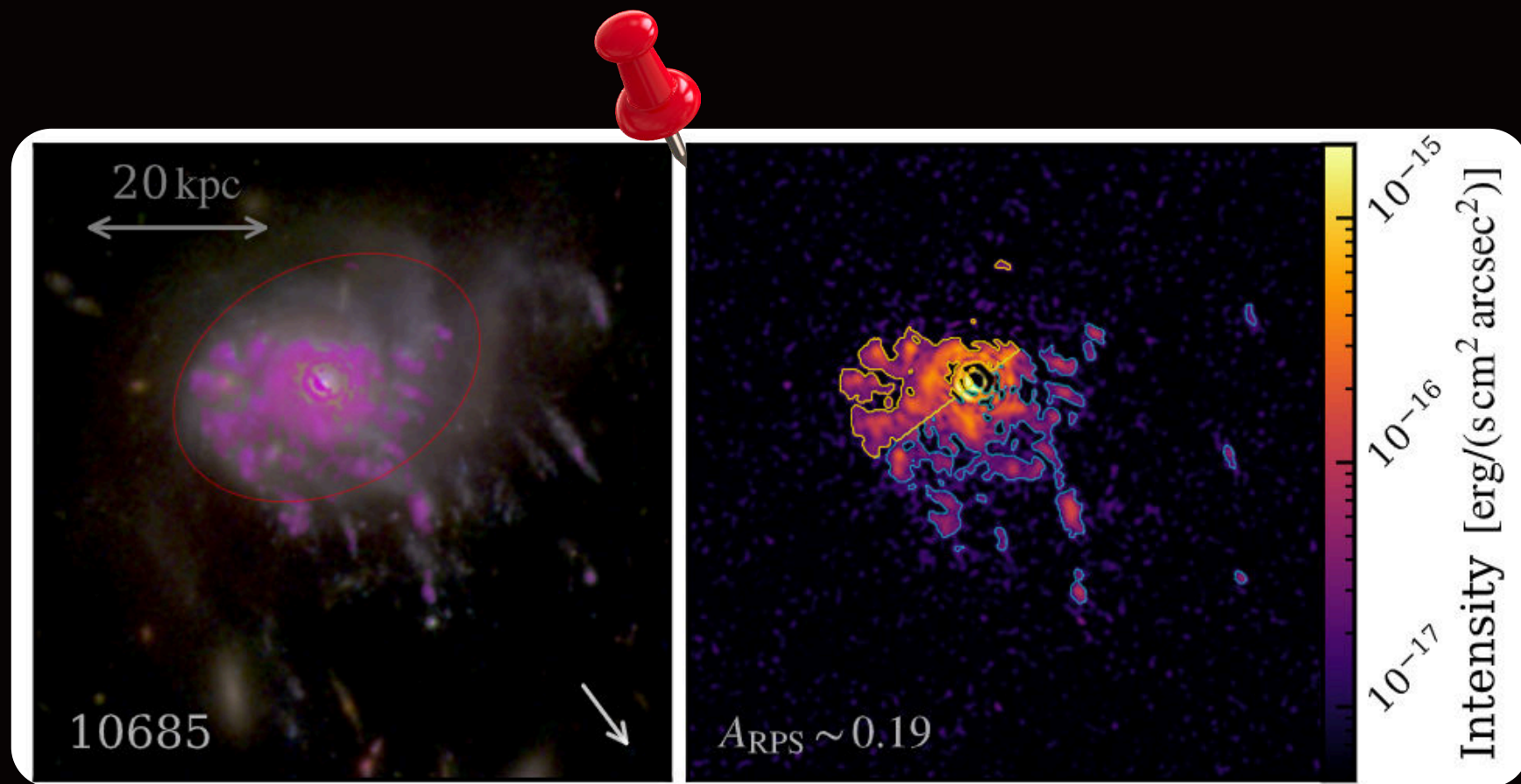


	CLUSTER	FIELD	
<u>STAR-FORMING</u>	59%	69% ($\pm 6\%$)	
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<u>QUIESCENT</u>	0%	5% ($\pm 2\%$)	

Benotto+2026c
(in prep.)



Benotto+2026a (A&A)



PAH emission in ram pressure stripped galaxies

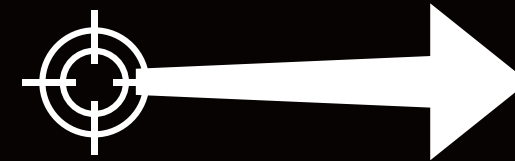
PAH molecules can be stripped together with the atomic gas

Conclusions

PAH emitting galaxies

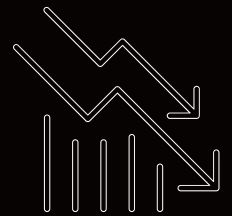


$5 \pm 1\%$ (clusters)
VS
 $44 \pm 4\%$ (fields)



The **PAH-emitting fraction** in clusters increases with cluster-centric distance and peak at $M = 10^{10} M_{\odot}$

Strong PAH suppression in cluster galaxies in the UVJ transition zone at 5σ

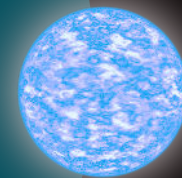


PAH EMISSION

found in stripped gas clumps

Environmental processing

has to be kept in mind to correctly calibrated the PAH emission as a SFR tracer



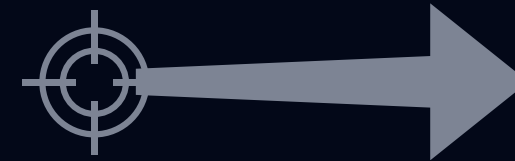
can have an impact on the PAH molecules directly



Conclusions

PAH emitting galaxies

$5 \pm 1\%$ (clusters)
VS
 $44 \pm 4\%$ (fields)



The **PAH-emitting fraction** in clusters increases with cluster-centric distance and peak at $M = 10^{10} M_{\odot}$

Strong PAH suppression in cluster galaxies in the UVJ transition zone at 5σ



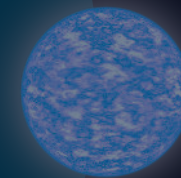
Thank you!

PAH EMISSION

found in stripped gas clumps

Environmental processing

has to be kept in mind to correctly calibrated the PAH emission as a SFR tracer



can have an impact on the PAH molecules directly

